



Making distributionally robust portfolios feasible in high dimension

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ABSTRACT

Robust estimation for modern portfolio selection on a large set of assets becomes more important due to the large deviation of empirical inference on big data. We propose a distributionally robust methodology for high-dimensional mean-variance portfolio problems, aiming to select an optimal conservative portfolio allocation by considering distributional uncertainty. With the help of factor structure, we extend the distributionally robust mean-variance problem investigated by Blanchet et al. (2022) to the high-dimensional scenario and transform it to a new penalized risk minimization problem. Furthermore, we propose a data-adaptive method to quantify both the uncertainty size and the lowest acceptable target return. Since the selection of these quantities requires knowledge of certain unknown population parameters, we further develop an estimation procedure, and establish its corresponding asymptotic consistency. Our Monte-Carlo simulation results show that the estimated uncertainty size and target return from the proposed procedure are very close to the corresponding oracle level, and the newly proposed robust portfolio achieves high out-of-sample Sharpe ratio. Finally, we conduct empirical studies based on the components of the S&P 500 index and the Russell 2000 index to demonstrate the superior return-risk performance of our proposed portfolio selection, in comparison with various existing strategies.

1. Introduction

The mean-variance portfolio selection problem proposed by Markowitz (1952) is one of the classical strategies in portfolio allocation. For a universe of p stocks, the Markowitz mean-variance model tries to choose an allocation w^* that minimizes portfolio risk while achieving the target expected return:

$$\begin{aligned} \min_w & w^\top \text{Var}_{\mathbb{P}^*}(r_t)w \\ \text{s.t. } & w^\top \mathbf{1}_p = 1, \quad \mathbb{E}_{\mathbb{P}^*}(w^\top r_t) \geq \bar{\alpha}, \end{aligned} \quad (1.1)$$

where w is the p -dimensional portfolio weight vector, r_t is a p -dimensional vector of assets return, $\mathbb{P}^*$ is the probability measure underlying the distribution of r_t , $\mathbb{E}_{\mathbb{P}^*}(\cdot)$ and $\text{Var}_{\mathbb{P}^*}(\cdot)$ are the expectation and variance under $\mathbb{P}^*$ respectively, $\bar{\alpha}$ is the target return, and $\mathbf{1}_p$ is a p -dimensional column vector of ones. However, in practical application, we do not always have prior information about population mean and covariance and hence rely instead on empirical versions of mean and covariance, which usually deviate

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significantly from the original versions (see, e.g., Michaud, 1989; Jagannathan and Ma, 2003; Ao et al., 2019). Uncertainty in data quality incurs statistical challenge for accurate inference.

These motivate the creation of the “robust” version of the Markowitz model, which aims to consider the influence of disparities between $\mathbb{P}^*$ and its empirical counterpart. Lobo and Boyd (2000) were the first to study the worst-case robust analysis with respect to second-order moment uncertainty in the framework of the Markowitz model. Goldfarb and Iyengar (2003) considered a vector/matrix distance for computing the uncertainty set. Pflug and Wozabal (2007) developed a Markowitz model with a Wasserstein distance-based ambiguity set and an additional value-at-risk type of constraint, but in their study, the uncertainty size is exogenous. Delage and Ye (2010) examined moment-based uncertainty regions in portfolio optimization without considering an optimal choice of the uncertainty set. Mohajerin Esfahani and Kuhn (2018) introduced representations for the worst-case expectations in a Wasserstein-based uncertainty set and further applied their approach to portfolio selection with various risk measures. A data-driven method for ambiguity size is also provided, but the choice deteriorates substantially with the increasing number of assets in portfolio selection.

Recently, Blanchet et al. (2022) (hereafter, BCZ) proposed a distributionally robust mean–variance problem by using the Wasserstein distance-measured uncertainty set, which performs very well in practical applications. They showed that the distributionally robust problem is equivalent to a regularized classical portfolio selection problem. Further, they provided data-driven methods for choosing uncertainty size and the lowest target portfolio return involved in their model. However, in high-dimensional situations, the selected uncertainty size is often overestimated, leading to overly conservative portfolio allocation due to the curse of dimensionality. Moreover, their theoretical results also fail under a high-dimensional framework. The selection of the uncertainty set is crucial for distributionally robust problems. Zhang et al. (2023) criticized the unnecessarily conservative allocation in BCZ, and proposed a new way to choose uncertainty size using the subspace Wasserstein distance. However, their method requires sample splitting, resulting in a relatively small sample size for each sub-block used to estimate the population mean and covariance.

In this paper, we consider a high-dimensional distributionally robust mean–variance problem, which extends the model of BCZ to high-dimensionality. To address the curse of dimensionality, we impose the factor structure for observed asset returns, and the model uncertainty is quantified for common factors via the Wasserstein distance. We believe that the uncertainty of asset returns can largely be explained by common factors that drive the primary co-movements of asset returns, especially during significant macroeconomic changes, such as the global financial crisis and the outbreak of COVID-19. Adding the factor structure for data in high-dimensional situations is quite popular and has become a frequently used workhorse (see, e.g., Fan et al., 2013; Ait-Sahalia and Xiu, 2017; Ding et al., 2021; Fan et al., 2024b, etc.). We first demonstrate that the distributionally robust optimization with factor-based asset return can also be reformed to a regularized risk-minimization problem. Similar to the BCZ model, our distributionally robust optimization in high-dimensional situations also includes two key parameters, δ and ρ , which depict the ambiguity set size and the lowest acceptable target return given uncertainty set, respectively. Second, we provide data-driven methods to determine the values of both δ and ρ . This is accomplished by adapting and extending the robust Wasserstein profile inference framework introduced by Blanchet et al. (2019). The core idea is to select δ and ρ such that the uncertainty set encompasses the true probability measure, and the feasible set of optimization includes the oracle mean–variance solution of (1.1), both at a specified confidence level. Third, since the optimal choice of these two parameters involve unknown population parameters, we provide related estimation approaches, as well as corresponding theoretical consistency under some regular conditions. It is worth noting that the asymptotic results of these estimations are established in the asymptotic regime where $p, T \rightarrow \infty$, and there is no conceptual upper bound on p/T in this paper. The implementation of the newly proposed portfolio is thus applicable even when $p > T$. Next, we conduct Monte Carlo simulations to evaluate the estimated optimal values of ρ and δ obtained from the proposed selection procedures. These estimated values are quite close to the unknown oracle versions. We also evaluate the performance of our high-dimensional distributionally robust portfolio, which exhibits higher Sharpe ratio compared to that of BCZ. Finally, we study empirical applications based on components of S & P 500 index and Russell 2000 index, finding that over a long-term investment from 2000 to 2023, the proposed high-dimensional distributionally robust portfolio exhibits an outstanding return–risk balance and achieves the highest Sharpe ratio in most scenarios, compared to various existing strategies, including the BCZ’s portfolio.

Our paper relates to the regularized portfolio allocation and high-dimensional portfolio allocation in finance. DeMiguel et al. (2009a) developed a general framework that solves the global minimum variance problem (GMVP) by incorporating the norm of the portfolio weight vector, such as the l_1 -norm or l_2 -norm, as a regularization term. Fan et al. (2012) investigated GMVP under gross-exposure constraint, where the l_1 -norm of the weight vector is required to be bounded. Based on GMVP, Fastrich et al. (2015) studied the construction of an optimal sparse portfolio using a weighted Lasso penalty and a non-convex SCAD penalty. Ho et al. (2015) proposed regularizing the mean–variance problem with a weighted elastic net penalty, which can be motivated by a robust reformulation of the mean–variance criterion that directly accounts for parameter uncertainty. Olivares-Nadal and DeMiguel (2018) regularized the mean–variance problem with p -norm transaction costs and showed equivalence between certain transaction costs and ellipsoidal robustification around the mean. Ding et al. (2021) developed a unified GMVP under statistical factor models in high-dimensional situations, with their approach also relying on properly integrating l_1 constraint on portfolio weights. Brodie et al. (2009) proposed adding l_1 -norm penalty to the mean–variance problem and reforming it as a constrained least squares regression problem. Ao et al. (2019) translated the mean–variance optimization into a novel unconstrained regression representation in a high-dimensional situation and theoretically proved that the regularized portfolio based on l_1 -norm regularization achieves the optimal mean–variance trade-off. However, their results rely on the Gaussian assumption and do not impose the budget constraint in (1.1). Li et al. (2022) argued that the strategy of Ao et al. (2019) would not apply to real large-scale portfolio allocation since the response variable involves the inverse of the sample covariance matrix. Consequently, they recommended applying a covariance matrix estimator based on the Fama–French factor model and performing portfolio allocation via a synthetic regression.

In addition, a regularized portfolio (with an appropriate penalized parameter) can be viewed as a nonlinear combination of an traditional portfolio and a portfolio with sufficiently strong regularization. Consequently, this paper also relates to literature on portfolio combination rules. Kan and Zhou (2007) developed an optimal three-fund rule that maximizes the expected out-of-sample utility in the case where a risk-free asset is available, their three-fund rule can be expressed as a combination of sample GMVP and the sample zero-investment portfolio (Kan et al., 2022). Tu and Zhou (2011) further improved their three-fund rule by combining it with the equally weighted portfolio. Kan et al. (2022) propose an optimal combining rule that explicitly takes into account estimation risk for the mean–variance problem in the case without a risk-free asset. The rule takes the form of optimally combining sample GMVP with a sample zero-investment portfolio. Yuan and Zhou (2024) also explored the combination of the equal-weighting rule with GMVP to maximize the Sharpe ratio and discussed scenarios in which the combination rule outperforms the equal-weighting strategy which is asymptotically optimal under their framework. Golosnoy and Okhrin (2007) considered a multivariate shrinkage estimator by shrinking portfolios with and without the riskless asset toward an arbitrary static portfolio. A similar technique was used by Frahm and Memmel (2010), who developed two feasible shrinkage estimators for the GMVP, which dominate the traditional estimator with respect to the out-of-sample variance of the portfolio return. Furthermore, Bodnar et al. (2018) derived a shrinkage estimator for the GMVP when the distribution of the asset returns is not prespecified and no market structure is imposed. Their proposed portfolio is also feasible when sample covariance matrix is singular ($p > T$). Bodnar et al. (2022) constructed a shrinkage estimator for the optimal portfolio weights obtained by shrinking the expected quadratic utility portfolio to an arbitrary target portfolio. The proposed strategy follows a mean–variance framework and is feasible when $p > T$. Lassance et al. (2024b) investigated how to best combine the sample mean–variance portfolio with the naive equally weighted portfolio to optimize out-of-sample performance. Garlappi et al. (2007) studied the portfolio selection with parameter and model uncertainty, showing that the optimal portfolio weights in the presence of ambiguity aversion can be viewed as a weighted combination of the mean–variance portfolio and the GMVP. Taking into parameter uncertainty, Lassance et al. (2024a) proposed a novel measure of portfolio robustness defined as the difference between out-of-sample utility mean and a multiple of out-of-sample utility standard deviation. They further developed an optimal shrinkage portfolio that optimally combines the sample mean–variance portfolio and sample GMVP, which maximizes this out-of-sample mean–risk criterion.

This paper is organized as follows. Section 2 introduces the basic high-dimensional distributionally robust problem based on a factor model. The corresponding dual problem is derived. Section 3 further studies the data-driven choice of required tuning parameters δ and ρ , the associated practical implementation of computing δ and ρ is given in Section 3.3. Section 4 presents an asymptotic theory. Section 5 presents simulation results to show the superiority of our proposed method. Section 6 develops an empirical application based on S & P 500 index and Russell 2000 index components. Section 7 concludes. The proof is included in the supplementary material.

In this paper, $\|A\|$, $\|A\|_1$, $\|A\|_F$, and $\|A\|_\infty$ denote the spectral norm, L_1 norm, Frobenius norm, and max norm of a matrix A , defined respectively by $\|A\| = \lambda_{\max}^{1/2}(A^T A)$, $\|A\|_1 = \max_j \sum_i |A_{ij}|$, $\|A\|_F = \text{tr}^{1/2}(A^T A)$ and $\|A\|_\infty = \max_i \sum_j |A_{ij}|$, $\lambda_{\max}(A)$ ($\lambda_{\min}(A)$) denotes the maximum (minimum) eigenvalues of a matrix A and $\text{tr}(A)$ is the trace of a matrix A . If A is a column vector, both $\|A\|$ and $\|A\|_F$ equal the Euclidean norm $\|A\|_2$, $\|A\|_i = (\sum_{j=1}^n |A_j|^i)^{1/i}$ for $i \geq 1$, and $\|A\|_\infty = \max_i |A_i|$. $\mathbb{P}^*$, $\mathbb{P}_T$, and $\mathbb{P}$ are true probability, empirical probability, and a general form of probability measures supported by $\mathbb{R}^x$ where x takes different values in different situations. $a_T \asymp b_T$ means that for some constants $c, C > 0$, $c < |a_T|/|b_T| < C$ for all T .

2. Methodology

2.1. MVP with distributional uncertainty

We begin by reviewing the distributionally robust mean–variance model investigated by Blanchet et al. (2022) (hereafter, BCZ). For a p -dimensional vector of asset return r_t , the distributionally robust mean–variance portfolio is given by solving:

$$\begin{aligned} \min_w \max_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} \{w^T \text{Var}_{\mathbb{P}}(r_t) w\} \\ \text{s.t. } w^T \mathbf{1}_p = 1, \min_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} \mathbb{E}_{\mathbb{P}}(w^T r_t) \geq \rho \end{aligned} \quad (2.1)$$

where w is portfolio weight, $\mathbb{E}_{\mathbb{P}}$ and $\text{Var}_{\mathbb{P}}$ take the expectation and variance under probability measure $\mathbb{P}$ respectively, $\mathbb{P}_T$ is the empirical probability measure, and $\mathcal{U}_\delta(\mathbb{P}_T) = \{\mathbb{P} : D_c(\mathbb{P}, \mathbb{P}_T) \leq \delta\}$ is the uncertainty set where $D_c(\cdot, \cdot)$ is the discrepancy between two probability measures based on a suitably defined Wasserstein distance

$$D_c(\mathbb{P}, \mathbb{Q}) := \inf \{ \mathbb{E}_{\pi} [c(U, V)] : \pi \in \mathcal{P}(\mathbb{R}^p \times \mathbb{R}^p), \pi_U = \mathbb{P}, \pi_V = \mathbb{Q} \}, \quad (2.2)$$

where $\mathbb{P}$ and $\mathbb{Q}$ are two probability measures supported on $\mathbb{R}^p$, $c(U, V)$ is a lower semi-continuous cost function such that $c(U, U) = 0$ for any $U \in \mathbb{R}^p$, $\mathcal{P}(\mathbb{R}^p \times \mathbb{R}^p)$ is the set of joint probability distributions π of (U, V) , and π_U, π_V denote the marginal distribution of U and V , respectively. Compared to the classic mean–variance problem (1.1), the distributionally robust formulation (2.1) introduces an artificial adversary $\mathbb{P}$ as a tool to account for the impact of the model uncertainty around the empirical distribution. The solution of (2.1) is robustly optimal because it minimizes the upper bound on portfolio risk. The uncertainty model involves two important parameters δ and ρ , which are the size of the ambiguity set and the lowest acceptable target portfolio return, respectively.

The min–max problem in (2.1) is complex and difficult to solve. If the cost function takes the form of $c(U, V) = \|U - V\|_i^2$ where $i \geq 1$ is fixed, Theorem 1 of BCZ shows that the dual problem of (2.1) is a regularized portfolio selection problem

$$\min_w \sqrt{w^T \text{Var}_{\mathbb{P}_T}(r_t) w} + \sqrt{\delta} \|w\|_q \quad (2.3)$$

$$\text{s.t. } w^\top \mathbf{1}_p = 1, \quad \mathbb{E}_{\mathbb{P}_T}(w^\top r_t) \geq \rho + \sqrt{\delta} \|w\|_q,$$

where $1/q + 1/\iota = 1$. The objective function and corresponding feasible region are convex with respect to w ; therefore, the new form (2.3) can be handled easily as a convex optimization problem.

2.2. Factor-structure MVP with uncertainty

We notice that the distributionally robust problem (2.3) measures the model uncertainty centered around the empirical covariance matrix $\text{Var}_{\mathbb{P}_T}(r_t)$ and empirical mean $\mathbb{E}_{\mathbb{P}_T}(r_t)$. However, it is well known that the sample covariance matrix performs poorly when p cannot be asymptotically negligible compared to the sample size T due to the curse of dimensionality. Furthermore, the sample mean is not good estimation in the portfolio allocation problem (see, e.g., Ledoit and Wolf, 2017; Ding et al., 2021; Fan et al., 2024b, investigating the minimum variance problem). If the center of uncertainty set deviates significantly from the population counterpart, the data adaptive uncertainty size¹ in BCZ tends to be quite large, leading to an unnecessarily conservative allocation. Consequently, the model (2.3) is not suitable for high-dimensional settings, where $\mathbb{P}_T$ often significantly deviates from the true one $\mathbb{P}^*$. In our empirical applications shown in Section 6, Section D of the supplementary material, and those from BCZ, the performance of the portfolio from (2.3) is very close to the equal weighting strategy, which is a special case when $\delta = \infty$.² Just as BCZ acknowledged, their approach fails when the number of stocks p is not small compared with the sample size T , and the empirical probability measure is not even consistent when $p > T$.

This paper investigates the distributionally robust mean–variance strategy available in high-dimensional situations. To deal with the curse of dimensionality, we impose the factor structure on the asset returns, which is common in the financial field (Fama and French, 1993, 2015; Fan et al., 2013). Following classic arbitrage pricing theory (Ross, 1976) and approximate factor model of Chamberlain and Rothschild (1983), the asset return can be modeled as³

$$r_{it} = b_i^\top F_t + e_{it}, \quad i = 1, \dots, p; \quad t = 1, \dots, T, \quad (2.4)$$

where r_{it} is the excess return of asset i at time t , F_t represents the common factors that drive the co-movement of asset returns with $\mu_f^* := \mathbb{E}_{\mathbb{P}^*}(F_t)$, b_i is a K -dimensional vector of factor loadings capturing the relationship between asset i and common factors, K is the number of common factors which is low dimensional and fixed, and e_{it} is the idiosyncratic error with zero mean and is uncorrelated with common factors. In matrix form, we can rewrite (2.4) as

$$R = BF + E,$$

where R is $p \times T$ excess return matrix, r_{it} is the i th row and j th column element of R , $B = (b_1, \dots, b_p)^\top$ is $p \times K$ factor loading matrix, $F = (F_1, \dots, F_T)$, and $E = (e_1, \dots, e_T)$ is $p \times T$ error matrix with $e_t = (e_{1t}, \dots, e_{pt})^\top$.

Based on factor structure, the population covariance matrix Σ_r^* can be decomposed as $B\Sigma_f^*B^\top + \Sigma_e^*$, where $\Sigma_f^* := \text{Var}_{\mathbb{P}^*}(F_t)$ and $\Sigma_e^* := \text{Var}_{\mathbb{P}^*}(e_t)$. As such, we consider the distributionally robust mean–variance problem available in high-dimensional setup as follows:

$$\begin{aligned} \min_w \left\{ \max_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} w^\top [B\text{Var}_{\mathbb{P}}(F_t)B^\top + \text{Var}_{\mathbb{P}}^*(e_t)] w \right\} \\ \text{s.t. } w^\top \mathbf{1}_p = 1, \quad \min_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} \mathbb{E}_{\mathbb{P}}(w^\top BF_t) \geq \rho, \end{aligned} \quad (2.5)$$

where the uncertainty set $\mathcal{U}_\delta(\mathbb{P}_T)$ is defined as $\mathcal{U}_\delta(\mathbb{P}_T) := \{\mathbb{P} : D_c^f(\mathbb{P}, \mathbb{P}_T) \leq \delta\}$, and $D_c^f(\mathbb{P}, \mathbb{P}_T)$ is the Wasserstein distance between the probability measure $\mathbb{P}$ and the empirical probability measure $\mathbb{P}_T$ of common factors, defined as

$$D_c^f(\mathbb{P}, \mathbb{P}_T) := \inf \left\{ \mathbb{E}_\pi (\|U - V\|_1) : \pi \in \mathcal{P}(\mathbb{R}^K \times \mathbb{R}^K), \pi_U = \mathbb{P}, \pi_V = \mathbb{P}_T \right\}, \quad (2.6)$$

where $\iota \geq 1$, $\mathcal{P}(\mathbb{R}^K \times \mathbb{R}^K)$ is the set of joint probability distributions π of (U, V) , and π_U, π_V are the corresponding marginal distributions.

Unlike model (2.1), which imposes uncertainty on the entire asset return r_t , we impose the ambiguity set centered around the empirical measure on common factors. The number of factors K is fixed, so it is reasonable for us to impose the uncertainty analysis on them directly. For the covariance matrix of the error term, a sparse structure is further assumed to address the curse of dimensionality. In addition, with the help of the factor structure, the target return constraint is simplified to $\min_{\mathbb{P}} w^\top B\mathbb{E}_{\mathbb{P}}(F_t) \geq \rho$ given that the error term does not contribute to the expected excess return. Intuitively, the newly proposed optimization problem (2.5) consists of performing a mean–variance optimization with uncertainty in the factor space, while simultaneously conducting a minimum-variance optimization in the remaining space. The solution to the optimization problem is obtained by jointly solving these two subproblems.

Similar to the model (2.1), the proposed optimization problem (2.5) also involves two key parameters δ and ρ . The parameter δ is generally interpreted as the power given to the adversary $\mathbb{P}$. The larger the value of δ , the more model ambiguity and the

¹ In BCZ, the selected value of δ ensures that the uncertainty set $\mathcal{U}_\delta$ includes the true probability measure at a given level, e.g., 95%.

² See Remark 1 for an intuitive interpretation.

³ Without loss of generality, and for notational simplicity, the factor model without intercept term is applied. We note that F_t can be rewritten as $\mu_f + \tilde{F}_t$ where $\tilde{F}_t$ has mean $\mathbb{E}_{\mathbb{P}^*}(F_t) - \mu_f$ with μ_f being an arbitrary K -dimensional column vector, and thus the factor model (2.4) is equivalent to $r_{it} = b_i^\top \mu_f + b_i^\top \tilde{F}_t + e_{it}$.

less important the information in the covariance matrix of F_t . The parameter ρ is typically interpreted as the lowest acceptable target return and, logically, should be selected according to the desired portfolio return level $\bar{\alpha}$. Notice that although the target portfolio return $\bar{\alpha}$ is often subjectively determined by investor's preferences, the choice of ρ must still be made with caution, as the feasible regions of problem (2.1) and (2.5) may be empty if the value of ρ is improperly specified.⁴ Clearly, choosing $\rho = \bar{\alpha}$ is overly aggressive, as the optimization problem requires the expected portfolio return in the worst case to exceed ρ , which often renders the problem infeasible.

The following [Theorem 1](#) reformulates the min-max problem (2.5) to its dual form, which is much easier to solve in practice.

Theorem 1. *The primal formulation given in (2.5) is equivalent to the following dual problem:*

$$\begin{aligned} \min_w & \left(\sqrt{w^\top B \text{Var}_{\mathbb{P}_T}(F_t) B^\top w} + \sqrt{\delta} \|B^\top w\|_q \right)^2 + w^\top \Sigma_e^* w \\ \text{s.t. } & w^\top \mathbf{1}_p = 1, w^\top B \mathbb{E}_{\mathbb{P}_T}(F_t) \geq \rho + \sqrt{\delta} \|B^\top w\|_q, \end{aligned} \quad (2.7)$$

where $1/i + 1/q = 1$.

The transformation is obtained based on a duality argument shown in the supplementary material. The objective function of (2.7) consists of the factor part (in the bracket) and the residual part. For the factor part, an additional regularization term $\sqrt{\delta} \|B^\top w\|_q$ is introduced by the uncertainty set imposed, and similar to the regularizations in portfolio allocation problems (see, e.g., [DeMiguel et al., 2009a](#); [Olivares-Nadal and DeMiguel, 2018](#)), the penalty $\sqrt{\delta} \|B^\top w\|_q$ provides robustness to the optimal portfolio from (2.7).

For the newly proposed mean-variance optimization (2.7) with uncertainty, when $\delta = 0$ (no ambiguity), it is clear that the optimization problem degenerates the mean-variance problem with factor model-based covariance matrix and mean as follows:

$$\begin{aligned} \min_w & w^\top \left(B \text{Var}_{\mathbb{P}_T}(F_t) B^\top + \Sigma_e^* \right) w \\ \text{s.t. } & w^\top \mathbf{1}_p = 1, w^\top B \mathbb{E}_{\mathbb{P}_T}(F_t) \geq \rho. \end{aligned}$$

For the case of $\delta > 0$, let $q = 2$ and denote $\hat{\Sigma}_f = \text{Var}_{\mathbb{P}_T}(F_t)$, then the robust optimization problem (2.7) (assuming no target return constraint) has the first-order condition (FOC) as follows

$$\left(B \hat{\Sigma}_f B^\top + \delta B B^\top + \Sigma_e^* \right) w + \sqrt{\delta} \left(\frac{\sqrt{w^\top B B^\top w}}{\sqrt{w^\top B \hat{\Sigma}_f B^\top w}} B \hat{\Sigma}_f B^\top w + \frac{\sqrt{w^\top B \hat{\Sigma}_f B^\top w}}{\sqrt{w^\top B B^\top w}} B B^\top w \right) = \frac{\mathcal{L}}{2} \mathbf{1}_p,$$

where $\mathcal{L}$ is the Lagrange multiplier. It is evident that the FOC is highly nonlinear with respect to weight w , and a closed-form solution is not easy to be obtained. Let $a_w := \sqrt{w^\top B B^\top w} / \sqrt{w^\top B \hat{\Sigma}_f B^\top w}$, then the FOC can be rewritten as

$$\left[\left(\delta + \sqrt{\delta}/a_w \right) B B^\top + \left(1 + \sqrt{\delta} a_w \right) B \hat{\Sigma}_f B^\top + \Sigma_e^* \right] w = \mathcal{L} \mathbf{1}_p / 2.$$

Notice that, both matrices $B B^\top$ and $B \hat{\Sigma}_f B^\top$ are low-rank, with their first K eigenvalues being positive and all remaining eigenvalues equal to zero. As such, under the assumption that Σ_e^* is positive definite, the optimal solution w^* satisfies $w^* = \Sigma_{r,w^*,\delta}^{-1} \mathcal{L}^* \mathbf{1}_p / 2 = \sum_{i=1}^p \gamma_i v_i v_i^\top \mathcal{L}^* \mathbf{1}_p / 2$, where $\mathcal{L}^*$ is the multiplier at optimality, $\Sigma_{r,w^*,\delta} := \left(\delta + \sqrt{\delta}/a_{w^*} \right) B B^\top + \left(1 + \sqrt{\delta} a_{w^*} \right) B \hat{\Sigma}_f B^\top + \Sigma_e^*$, and γ_i, v_i are the i th largest eigenvalue and the corresponding eigenvector of $\Sigma_{r,w^*,\delta}$, respectively. As a consequence, it is clear that Σ_e^* plays an important role in determining w^* , since the largest $(p - K)$ eigenvalues γ_i depend on the smallest $(p - K)$ eigenvalues of Σ_e^* .

For the case of $\delta \rightarrow \infty$, for ease of notation, let us denote $\theta_{\delta,1} = \delta + \sqrt{\delta}/a_{w^*}$, $\theta_{\delta,2} = 1 + \sqrt{\delta} a_{w^*}$, and $M_\delta = \theta_{\delta,1} \mathbf{I}_K + \theta_{\delta,2} \hat{\Sigma}_f$. Since $\Sigma_{r,w^*,\delta}$ is the sum of a low-rank matrix and a positive definite matrix, we can apply the famous Sherman-Morrison-Woodbury formula and derive that

$$\Sigma_{r,w^*,\delta}^{-1} = \Sigma_e^{*-1} - \Sigma_e^{*-1} B (M_\delta^{-1} + B^\top \Sigma_e^{*-1} B)^{-1} B^\top \Sigma_e^{*-1}. \quad (2.8)$$

For M_δ^{-1} , note that we can apply the eigenvalue decomposition $\hat{\Sigma}_f = Q Y_f Q^\top$, where Q is an orthogonal matrix. It then follows that $M_\delta^{-1} = (\theta_{\delta,1} \mathbf{I}_K + \theta_{\delta,2} Q Y_f Q^\top)^{-1} = Q (\theta_{\delta,1} \mathbf{I}_K + \theta_{\delta,2} Y_f)^{-1} Q^\top = Q \left(\frac{1}{\theta_{\delta,1} + \theta_{\delta,2} Y_{f,11}}, \dots, \frac{1}{\theta_{\delta,1} + \theta_{\delta,2} Y_{f,KK}} \right) Q^\top$. As $\delta \rightarrow \infty$, it is clear that $\theta_{\delta,1} + \theta_{\delta,2} Y_{f,ii} = \delta + o(1)$ for $i = 1, \dots, K$, therefore, $\Sigma_{r,w^*,\delta}^{-1}$ is asymptotically equivalent to $(\delta B B^\top + \Sigma_e^*)^{-1}$ as $\delta \rightarrow \infty$ according to (2.8). Define $\Omega_{r,\infty}^* := \lim_{\delta \rightarrow \infty} (\delta B B^\top + \Sigma_e^*)^{-1}$, then at optimality, $w^* = \Omega_{r,\infty}^* \mathcal{L}^* \mathbf{1}_p / 2$ is exact the FOC of minimum variance problem with covariance matrix $\Omega_{r,\infty}^{*-1}$. Consequently, as $\delta \rightarrow \infty$, our robust optimal portfolio (assuming no target return constraint) converges to the minimum variance portfolio with covariance matrix $\Omega_{r,\infty}^{*-1}$.

In what follows, we focus on the case of $q = 2$, as this setting allows us to derive explicit expressions for some of the results presented later. It is also relatively simple and may be more favored by practitioners. We note that in the empirical analysis of [BCZ](#), they discussed the case of $q = 1$, which resulted in some volatile results (see their discussions in Section 5.3). Additionally, although the cost function is chosen as $\|U - V\|_t^2$ here, the reformulation in [Theorem 1](#) holds for any cost function of the form $c(U, V) = \|U - V\|^2$ where $\|\cdot\|$ is any given norm with a suitable dual, see more arguments in [BCZ](#).

⁴ See more in [Remark 2](#) and [Simulation 5.4](#).

Remark 1 (Intuition for Limiting Portfolio). We provide some intuition for the limiting portfolio when $\delta \rightarrow \infty$. For robust optimization of BCZ, when uncertainty size is sufficiently large, the uncertainty set $\mathcal{U}_\delta(\mathbb{P}_T)$ encompass almost all possible scenarios, implying that the information in $\mathbb{P}_T$ derived from the data becomes less effective. As a result, all assets are treated as equivalent in BCZ, and the optimal strategy reduces to allocating equal weights across assets, since no asset can be reliably distinguished from another in terms of expected performance. For our newly proposed optimization problem, it consists of a mean–variance optimization in the factor space under uncertainty, combined with a minimum variance optimization in the remaining space. Following the same interpretation, as uncertainty size becomes sufficiently large, all common factors become approximately indistinguishable. However, the information contained in error covariance matrix remain influential. Consequently, the optimal allocation as $\delta \rightarrow \infty$ does not converge to the equally weighted portfolio unless the error covariance matrix is the identity matrix.

Remark 2 (Empty Feasible Region). It is worth noting that both the constraints $E_{\mathbb{P}_T}(w^\top r_t) \geq \rho + \sqrt{\delta}\|w\|_q$ in (2.3) of BCZ and $w^\top BE_{\mathbb{P}_T}(F_t) \geq \rho + \sqrt{\delta}\|B^\top w\|_q$ in our new optimization (2.7) may result in empty feasible regions. For a simple example, consider the case of $p = 2$ and $E_{\mathbb{P}_T}(r_t) = (a, a)^\top$. In this case, the target return constraint of BCZ becomes $a(w_1 + w_2) = a \geq \rho + \sqrt{\delta}\|w\|_q$. It is evident that if $\rho \geq a$, no w satisfy this target return constraint. The larger the ambiguity size, the smaller the feasible region of w , since the target return constraint requires the expected portfolio return to exceed a threshold ρ in the worst scenario. Consequently, it is important to first check whether the feasible region is empty before solving both the dual formulation (2.3) and (2.7). Fortunately, the data adaptive choice of ρ , as introduced in the subsequent Section 3.2, automatically ensures a non-empty set with high probability, simplifying the practical application of the proposed strategy. Further discussion can be found in Simulation 5.4.

Remark 3 (Optimal Allocation Given Specific $\bar{a}$). For a mean–variance investor with a specific target return $\bar{a}$, if he/she knows the information of true covariance matrix and true mean return, the optimal portfolio is the solution w_{mv}^* of problem (1.1). In practical applications without population information, a mean–variance investor acknowledging uncertainty, would prefer to construct an optimal portfolio from (2.5). A natural question that arises is, for a specific target return $\bar{a}$, what is the corresponding optimal ρ ? Similar to the idea of BCZ, we determine the optimal value of ρ based on the criterion that it ensures the feasible region contains the true mean–variance portfolio w_{mv}^* calculated at target return $\bar{a}$ with high probability. Based on this criterion, the optimal ρ is given by (3.12). As a result, the optimal portfolio allocation given $\bar{a}$ is the solution of the new optimization problem, with ρ selected from our data-adaptive method introduced in the following Section 3.2. Note that, a large positive value for ρ can be assigned, but it may lead to no solution. In our simulation, simply setting $\rho = \bar{a}$ leads to no available solution, see Remark 2. Of course, investors can also arbitrarily assign a strongly negative value to ρ , which ensures the availability of a solution, but in this case, the target return constraint tends to become slack and useless.

At the end of this section, we further provide some discussions on uncertainty learning for the error component. As the error component is of high-dimensional data structure, the traditional Wasserstein distance in (2.6) is not directly applicable due to curse of dimensionality. To deal with this challenge, a modified (e.g. regularized) Wasserstein distance should be utilized in the uncertainty quantification for error component. In this paper, we majorly contribute to uncertainty learning for the low-rank structure (i.e. common factors) of high-dimensional data. As further study, we will extend the distributionally robust estimation to the high-dimensional error component by modifying the Wasserstein distance.

However, we also analyze the portfolio selection if applying the current methodology on the error component directly. Details are in Section C of the supplementary material. For convenience, we present a brief summary here. By noting that the common factors and error term are assumed to be uncorrelated, it is sufficient to consider the following maximization problem:

$$\max_{\mathbb{P} \in \mathcal{U}_{\delta_e}(\mathbb{P}_T)} \{w^\top E_{\mathbb{P}}(e_t e_t^\top) w\}, \quad (2.9)$$

where $\mathcal{U}_{\delta_e}$ and δ_e represent the uncertainty set and the corresponding uncertainty size, respectively. Our Proposition C.1 shows that the optimal value of (2.9) is given by $\left(\sqrt{w^\top \text{Var}_{\mathbb{P}_T}(e_t) w} + \sqrt{\delta_e}\|w\|_q\right)^2$. We note that the above formulation is derived entirely via a duality argument, which relies solely on algebraic calculations and is irrelevant to statistical inference. Similar to Theorem 1, the impact of imposing uncertainty set can be translated into a penalization term. As a result, based on factor model, the distributionally robust mean–variance optimization, which considers both uncertainty for the common factors and the error term, can be expressed as follows:

$$\begin{aligned} \min_w & \left(\sqrt{w^\top B \text{Var}_{\mathbb{P}_T}(F_t) B^\top w} + \sqrt{\delta}\|B^\top w\|_q \right)^2 + \left(\sqrt{w^\top \text{Var}_{\mathbb{P}_T}(e_t) w} + \sqrt{\delta_e}\|w\|_q \right)^2 \\ \text{s.t. } & w^\top \mathbf{1}_p = 1, w^\top BE_{\mathbb{P}_T}(F_t) \geq \rho + \sqrt{\delta}\|B^\top w\|_q. \end{aligned} \quad (2.10)$$

However, if we further consider the choice of uncertainty size δ_e within our framework, using the logic introduced in the following Section 3.2, then under certain regularization conditions and based on techniques from random matrix theory, Theorem C.1 in Section C of the supplementary material demonstrates that the data-adaptive uncertainty size for the error term tends to positive infinity at a rate of order p as $T, p \rightarrow \infty$. As a result, if uncertainty learning is also applied to the error term with the data-driven value of δ_e used, the optimal solution to the problem in (2.10) approximates to the equal weighting allocation, indicating that both the information of common factors and error term are useless. As well known, the information of error term plays an important role in asset allocation, particularly when factor number K is not large. Consequently, we focus on the problem (2.7), which preserves the information of the error covariance matrix in the framework of our study.

3. Uncertainty quantification

In the first two Sections 3.1 and 3.2, we provide guidance on the choices of the ambiguity size δ , and the lowest acceptable target portfolio return ρ . The choice of both parameters should be informed by the data based on some statistical principles rather than being arbitrarily exogenous.⁵ Note that the studies in Sections 3.1 and 3.2 are oracle. In Section 3.3, we further provide the estimation procedure for all related unknown population parameters during the selections of both δ and ρ .

3.1. Choice of δ

The choice of uncertainty size δ is crucial for problem (2.5). When δ is too small, the effect of robustification becomes negligible; in the extreme case where $\delta = 0$, the problem reduces to the classical mean–variance optimization without accounting for distributional uncertainty. On the other hand, if δ is excessively large, the uncertainty set $\mathcal{U}_\delta(\mathbb{P}_T)$ becomes overly conservative, encompassing a vast range of probability measures. In such cases, the empirical distribution $\mathbb{P}_T$ derived from the data contributes little to the optimization. Furthermore, the feasible region of our optimization problem may also become overly restrictive for a given ρ , since the target return constraint requires the expected portfolio return to exceed ρ under the worst scenario. See more discussions in Remarks 1 and 2.

In this paper, the logic of choosing the value for δ is that it should be large enough such that the uncertainty set $\mathcal{U}_\delta(\mathbb{P}_T)$ contains the true probability measure $\mathbb{P}^*$ with a given confidence level. This idea was first introduced by Blanchet et al. (2019). The construction of the confidence interval poses a challenging problem. Here, we design a minimum-variance framework to construct a new statistic for detecting the optimal value of δ .

Let us first introduce the classic minimum variance problem (3.1) based on common factors F_t under the probability measure $\mathbb{P}$, which serves as the mapping relationship:

$$\min_{\phi} \phi^\top \text{Var}_{\mathbb{P}}(F_t) \phi, \quad \text{s.t.} \quad \phi^\top \mathbf{1}_K = 1, \quad (3.1)$$

where $\text{Var}_{\mathbb{P}}(F_t) = (\mathbb{E}_{\mathbb{P}}(F_t F_t^\top) - \mathbb{E}_{\mathbb{P}}(F_t) \mathbb{E}_{\mathbb{P}}(F_t^\top))$ is the variance–covariance matrix of F_t under probability measure $\mathbb{P}$.

Remark 4. Unlike the choosing procedure in BCZ, which uses the mean–variance problem to select the value of δ ,⁶ we choose the minimum-variance problem of common factors because it does not require an additional exogenous target return for the factor portfolio. It is clear that the target return for the factor-based optimization is logically related to $\bar{\alpha}$, but differs from $\bar{\alpha}$ due to the presence of the factor loading.

In the following, several technical assumptions are imposed to develop the rule for choosing δ .

Assumption 1.

- (a) The common factor $\{F_t\}_{t=1}^T$ is a stationary and ergodic process satisfying $\mathbb{E}_{\mathbb{P}^*}(\|F_t\|_2^4) < \infty$ for all t .
- (b) For function $g(x) = c_1 x + c_2 x x^\top v$, where c_1, c_2 are some constants, v is a constant vector, the limit $V_g := \lim_{T \rightarrow \infty} \text{Var}_{\mathbb{P}^*} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T g(F_t) \right)$ exists, and the central limit theorem holds:

$$\frac{1}{\sqrt{T}} \sum_{t=1}^T (g(F_t) - \mathbb{E}_{\mathbb{P}^*}(g(F_t))) \xrightarrow{d} Z_0 := N(0, V_g).$$

Assumption 2. For any deterministic matrix $\Lambda \in \mathbb{R}^{K \times K}$ and any deterministic vector $\gamma \in \mathbb{R}^K$ such that either $\Lambda \neq 0$ or $\gamma \neq 0$,

$$\mathbb{P}^* (\|\Lambda F_t + \gamma\|_2 > 0) > 0.$$

Assumption 3. Both Σ_f^* and $\tilde{\Sigma}_f^* = \mathbb{E}_{\mathbb{P}^*}(F_t F_t^\top)$ are positive definite.

Assumption 1(a) requires common factor F_t to be a stationary and ergodic process with mild moment conditions. Assumption 1(b) assumes a central limit theorem (CLT) for the sum of $g(F_t)$. When $c_1 = 1, c_2 = 0$, it indicates the CLT for a stationary factor; When $c_1 = 0, c_2 = 1, v = \mathbf{I}_K$, it implies the CLT for the sample covariance matrix, which is common in random matrix theory. Clearly, our Assumption 1 can be indicated by Assumption 1 of BCZ. Assumption 2 holds by assuming that F_t has a density, a similar condition

⁵ Note that the value of ρ logically depends on the target return $\bar{\alpha}$, which is specified by investors' preferences and is exogenous. However, the manner in which ρ depends on $\bar{\alpha}$ must be carefully determined, as an arbitrarily chosen value of ρ may lead to an empty feasible region and exclude the true optimal portfolio from consideration, as discussed in Remarks 2 and 3. Therefore, it is necessary to develop a data-adaptive procedure to guide the appropriate selection of ρ based on the user-specified target return $\bar{\alpha}$.

⁶ In their study, they choose the uncertainty size under a mean–variance framework:

$$\min_{\phi} \phi^\top \mathbb{E}_{\mathbb{P}}(r_t r_t^\top) \phi, \quad \text{s.t.} \quad \phi^\top \mathbf{1}_p = 1, \quad \phi^\top \mathbb{E}_{\mathbb{P}}(r_t) = \rho.$$

can be found in Blanchet et al. (2019) and BCZ. Assumption 3 ensures that problem (3.1) has a unique solution when $\mathbb{P} = \mathbb{P}^*$, and therefore, there exists a unique Lagrange multiplier λ_f^* such that

$$2(\mathbb{E}_{\mathbb{P}^*}(F_t F_t^\top) - \mathbb{E}_{\mathbb{P}^*}(F_t) \mathbb{E}_{\mathbb{P}^*}(F_t^\top)) \phi_f^* - \lambda_f^* \mathbf{1}_K = 0, \quad (3.2)$$

where $\phi_f^* := \phi_{f, \mathbb{P}^*}$, and $\phi_{f, \mathbb{P}}$ is the solution to minimum variance problem (3.1) under probability measure $\mathbb{P}$.

Let $\Phi_{f, \mathbb{P}}$ be a set collecting all solutions $\phi_{f, \mathbb{P}}$.⁷ Based on Assumption 3, it is clear that $\Phi_{f, \mathbb{P}^*} = \{\phi_f^*\}$. Now we can further define that

$$\Gamma_\delta(\mathbb{P}_T) = \bigcup_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} \Phi_{f, \mathbb{P}}$$

collects all solutions of problem (3.1) with $\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)$. Thus, the true probability measure $\mathbb{P}^*$ being included in $\mathcal{U}_\delta(\mathbb{P}_T)$ is equivalent to ϕ_f^* being included in $\Gamma_\delta(\mathbb{P}_T)$. As such, $\Gamma_\delta(\mathbb{P}_T)$ can be regarded as a natural confidence region for ϕ_f^* , and then, we can choose δ_T^* such that ϕ_f^* belongs to this region with a pre-specified confidence level. Specifically, we define

$$\delta_T^* = \min\{\delta > 0, \quad \mathbb{P}^*(\phi_f^* \in \Gamma_\delta(\mathbb{P}_T)) \geq 1 - \alpha_0\},$$

where $1 - \alpha_0$ denotes a given confidence level, for example, typically 95%.

However, it is still difficult to compute δ_T^* directly by its definition. Therefore, we follow the idea of BCZ and apply the robust Wasserstein profile function (RWP) to provide a simpler representation. Since our uncertainty set is imposed on the common factor rather than on the return itself, and we use the minimum variance problem as the framework, the analysis differs from that presented in BCZ. Before defining RWP, we first note that for any $\phi \in \Gamma_\delta(\mathbb{P}_T)$, there exists $\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)$ and Lagrange multiplier $\lambda_f \in (-\infty, \infty)$ such that

$$2(\mathbb{E}_{\mathbb{P}}(F_t F_t^\top) - \mathbb{E}_{\mathbb{P}}(F_t) \mathbb{E}_{\mathbb{P}}(F_t^\top)) \phi = \lambda_f \mathbf{1}_K. \quad (3.3)$$

Then, the robust Wasserstein profile function can be defined as follows:

$$\overline{\mathcal{R}}_T(\phi, \lambda_f, \mu, \Sigma) := \inf \{D_c^f(\mathbb{P}, \mathbb{P}_T) : \lambda_f \mathbf{1}_K = 2(\Sigma - \mu \mu^\top) \phi, \mu = \mathbb{E}_{\mathbb{P}}(F_t), \Sigma = \mathbb{E}_{\mathbb{P}}(F_t F_t^\top)\} \quad (3.4)$$

for $(\phi, \lambda_f, \mu, \Sigma) \in \mathbb{R}^K \times \mathbb{R} \times \mathbb{R}^K \times \mathbb{S}_+^{K \times K}$, where $\mathbb{S}_+^{K \times K}$ is the set of all the symmetric positively semi-definite matrices, $D_c^f(\mathbb{P}, \mathbb{P}_T)$ is the Wasserstein distance defined in (2.6). Further, define

$$\overline{\mathcal{R}}_T^*(\phi_f^*) := \inf_{\lambda_f \in \mathbb{R}, \mu \in \mathbb{R}^K, \Sigma \in \mathbb{S}_+^{K \times K}} \overline{\mathcal{R}}_T(\phi_f^*, \lambda_f, \mu, \Sigma), \quad (3.5)$$

we can observe that the events

$$\{\phi_f^* \in \Gamma_\delta(\mathbb{P}_T)\} \Rightarrow \{\overline{\mathcal{R}}_T^*(\phi_f^*) \leq \delta\}, \quad (3.6)$$

and for any given $\epsilon > 0$,

$$\overline{\mathcal{R}}_T^*(\phi_f^*) \leq \delta + \epsilon \Rightarrow \phi_f^* \in \Gamma_\delta(\mathbb{P}_T). \quad (3.7)$$

Let us define

$$\tilde{\delta}_T^* = \inf \left\{ \delta > 0, \quad \mathbb{P}^*(\overline{\mathcal{R}}_T^*(\phi_f^*) \leq \delta) \geq 1 - \alpha_0 \right\}.$$

Then, it follows from (3.6) and (3.7) that $\tilde{\delta}_T^* \leq \delta_T^* \leq \tilde{\delta}_T^* + \epsilon$ for arbitrary $\epsilon > 0$, as such, we obtain $\delta_T^* = \tilde{\delta}_T^*$. As a result, δ_T^* is given by the $(1 - \alpha_0)$ quantile of the distribution of $\overline{\mathcal{R}}_T^*(\phi_f^*)$.

In practice, it is difficult to handle $\overline{\mathcal{R}}_T^*(\phi_f^*)$ since it is a minimization problem in terms of the mean and positive semi-definite covariance. As an alternative, we consider a statistic that serves as an upper bound for $\overline{\mathcal{R}}_T^*(\phi_f^*)$, thereby providing the upper bound for δ_T^* . Specifically, we can define an upper bound $\mathcal{R}_T := \overline{\mathcal{R}}_T(\phi_f^*, \lambda_f^*, \mu_f^*, \tilde{\Sigma}_f^*)$. It is clear that $\mathcal{R}_T \geq \overline{\mathcal{R}}_T^*(\phi_f^*)$, and therefore,

$$\mathcal{R}_T \leq \delta \Rightarrow \overline{\mathcal{R}}_T^*(\phi_f^*) \leq \delta.$$

Then, we can further define the upper bound of δ_T^* as follows:

$$\bar{\delta}_T^* := \inf \left\{ \delta \geq 0 : \mathbb{P}^*(\mathcal{R}_T \leq \delta) \geq 1 - \alpha_0 \right\}. \quad (3.8)$$

Based on (3.8), we can use the upper bound $\bar{\delta}_T^*$, which is the upper α_0 quantile of term $\mathcal{R}_T$, as the choice of δ . Note that, with the choice of λ_f^* , μ_f^* , and $\tilde{\Sigma}_f^*$, the condition $\lambda_f \mathbf{1}_K = 2(\Sigma - \mu_f \mu_f^\top) \phi_f^*$ in $\overline{\mathcal{R}}_T$ can be naturally satisfied. As a result, the value of $\mathcal{R}_T$ is equal to

$$\mathcal{R}_T = \inf \left\{ D_c^f(\mathbb{P}, \mathbb{P}_T) : \mathbb{E}_{\mathbb{P}}(F_t F_t^\top) = \tilde{\Sigma}_f^*, \mathbb{E}_{\mathbb{P}}(F_t) = \mu_f^* \right\}.$$

Clearly, if we can obtain the limit of $\mathcal{R}_T$, the corresponding $1 - \alpha_0$ quantile of the limiting distribution is the selected value of uncertainty size δ . The following Theorem 2 provides the asymptotic properties of the term $\mathcal{R}_T$.

⁷ The solution for problem (3.1) under probability measure $\mathbb{P}$ may be not unique since $\text{Var}_{\mathbb{P}}(F_t)$ may be positive semi-definite.

Theorem 2. Suppose [Assumptions 1–3](#) hold true, if $q = 2$, then we have

$$T\mathcal{R}_T \xrightarrow{d} L_0 = \frac{\|Z_0\|_2^2}{4 \left(1 - \mu_f^{*\top} \tilde{\Sigma}_f^{*-1} \mu_f^*\right)} + \frac{1}{4} \sum_{i=1}^K \|\tilde{\Sigma}_f^{*-1/2} Z_{2,i}\|_2^2, \quad (3.9)$$

where $Z_0 \sim N(0, V_{g1})$ and $Z_{2,i} \sim N(0, V_{f,i})$, V_{g1} and $V_{f,i}$ are defined as V_g in [Assumption 1](#), where the function $g(x)$ takes the form $g_1(x) = -x + xx^\top \tilde{\Sigma}_f^{*-1} \mu_f^*$ for V_{g1} and $g_{f,i}(x) = xx^\top I_i$ for $V_{f,i}$, respectively, and I_i is a K -dimensional column vector whose i th element is 1 and all other elements are 0.

The limit L_0 contains only elements characterized by the fixed K -dimension and is free of the number of assets p , making it suitable for high-dimensional situations. Furthermore, compared to the mean–variance counterpart derived by [BCZ](#) when $q = 2$, our limit L_0 includes an additional term $\frac{1}{4} \sum_{i=1}^K \|\tilde{\Sigma}_f^{*-1/2} Z_{2,i}\|_2^2$ which is introduced by the term $\text{tr}(\Lambda(S - \Sigma))$ with $\Sigma = \tilde{\Sigma}_f^*$ where $\Lambda \in \mathbb{R}^{K \times K}$ is an optimizing variable and $S = E_{\mathbb{P}_T}(F_t F_t^\top)$. In [BCZ](#), the corresponding term $\text{tr}(\Lambda(S_r - \Sigma))$ is eliminated by letting $\Sigma = S_r$ where $S_r = E_{\mathbb{P}_T}(r_t r_t^\top)$, but the cost is that the Z_0 in their Theorem 2 is considerably more complicated than ours and involves the unknown optimal mean–variance portfolio, which is difficult to estimate consistently when applied to common factors.

3.2. Choice of ρ

Once δ has been chosen, the next step is to choose the value of ρ . Given δ , if ρ is chosen too large, the feasible region of problem (2.5) may become too small or even empty. If ρ is chosen as too small, the target return constraint may become slack and inactive. We highlight again that the value of ρ logically depends on the investor's own preference $\bar{\alpha}$, and the manner in which ρ depends on $\bar{\alpha}$ should be determined with care.

In this paper, we select the value for ρ such that we do not rule out the inclusion $w_{mv}^*(\bar{\alpha}) \in \mathcal{F}_{\delta,\rho}(T)$ with a given confidence level $1 - \epsilon$, where $w_{mv}^*(\bar{\alpha})$ is the optimal solution of the classic mean–variance problem (1.1) under $\mathbb{P}^*$ with target return $\bar{\alpha}$ and

$$\mathcal{F}_{\delta,\rho}(T) = \left\{ w \in \mathbb{R}^p : w^\top \mathbf{1}_p = 1, \min_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} E_{\mathbb{P}}(w^\top B F_t) \geq \rho \right\}$$

is the feasible set of our robust optimization problem (2.5). By applying the proposition A.1.⁸ of [BCZ](#), it follows that $w_{mv}^* \in \mathcal{F}_{\delta,\rho}(T)$ if and only if

$$w_{mv}^{*\top} B E_{\mathbb{P}_T}(F_t) - \sqrt{\delta} \|B^\top w_{mv}^*\|_q \geq \rho.$$

It is evident that if w_{mv}^* is included in the feasible region $\mathcal{F}_{\delta,\rho}(T)$, it should satisfy the conditions required in $\mathcal{F}_{\delta,\rho}(T)$. Notice that $\bar{\alpha} = w_{mv}^{*\top} E_{\mathbb{P}^*}(r_t) = w_{mv}^{*\top} B \mu_f^*$, thus we can rewrite the previous inequality as

$$w_{mv}^{*\top} B \left(E_{\mathbb{P}_T}(F_t) - E_{\mathbb{P}^*}(F_t) \right) \geq \rho - \bar{\alpha} + \sqrt{\delta} \|B^\top w_{mv}^*\|_q. \quad (3.10)$$

Based on [Assumption 1\(b\)](#), the left hand side of (3.10) has the following limiting behavior:

$$(w_{mv}^{*\top} B) \frac{1}{\sqrt{T}} \sum_{t=1}^T \left(F_t - \mu_f^* \right) \xrightarrow{d} Z_1 \sim N(0, w_{mv}^{*\top} B V_F B^\top w_{mv}^*), \quad (3.11)$$

where V_F is long-run covariance matrix of common factors defined in [Assumption 1](#) with $g(x) = x$. As a result, we can choose ρ sufficiently negative such that the inequality (3.10) holds (i.e., true optimal mean–variance portfolio is included in the feasible region) at a given confidence level $1 - \epsilon$. Let $\mathcal{A}_\epsilon$ denote the ϵ quantile of normal variate Z_1 , based on (3.10) and (3.11), the value of ρ is selected by

$$\rho_{hd} = \bar{\alpha} - \sqrt{\delta} \|B^\top w_{mv}^*\|_q + \mathcal{A}_\epsilon / \sqrt{T}. \quad (3.12)$$

From the formula (3.12), we observe the following: First, the selected ρ_{hd} is related to the target portfolio return $\bar{\alpha}$ and the uncertainty size, which aligns well with intuition. Second, the selected ρ_{hd} is generally smaller than the target return $\bar{\alpha}$,⁹ which is reasonable, as we consider uncertainty learning, and ρ represents the lowest acceptable target return in the worst scenario. The larger uncertainty size, the lower optimal selected value ρ_{hd} . Finally, and most importantly, the selected ρ_{hd} ensures that the feasible region of the newly proposed robust optimization is non-empty with high probability, as the criterion we use guarantees that the feasible region $\mathcal{F}_{\delta,\rho}$ at least (with high probability $1 - \epsilon$) contains the true mean–variance portfolio w_{mv}^* . This observation is particularly appealing in practical application because the solution set of the robust optimization problem may be empty, see more discussion in [Remark 2](#). With the selected ρ_{hd} , it is unnecessary to verify whether the feasible region is empty before solving the proposed optimization problem, which simplifies practical implementation.

⁸ Proposition A.1 of [BCZ](#) says that, for $c(u, v) = \|u - v\|_2^2$, we have

$$\min_{\mathbb{P} \in \mathcal{U}_\delta(\mathbb{P}_T)} E_{\mathbb{P}}(w^\top r_t) = E_{\mathbb{P}_T}(w^\top r_t) - \sqrt{\delta} \|w\|_q,$$

where $1/i + 1/q = 1$.

⁹ The significance level ϵ often takes a value less than 0.5.

3.3. Implementation

In practice, the population quantities are generally unknown. As replacements, consistent estimations of the optimal choices for both δ and ρ are required, which further necessitate the estimations of V_{g1} , $V_{f,i}$, V_F , $\mu_f^{*\top} \tilde{\Sigma}_f^{*-1} \mu_f^*$, $w_{mv}^{*\top} B V_F B^\top w_{mv}^*$ and $\|B^\top w_{mv}^*\|_2$.

For ease of reading, we first present the estimation algorithm that outlines the detailed procedure for estimating the selected δ and ρ . Note that the choices of several tuning parameters, such as the kernel function, truncation parameter, shrinkage function, and so on, are provided in detail in the subsequent simulation and empirical sections.

Algorithm 1 Estimation procedure for the data-adaptive δ and ρ

- (1). Estimate common factors and factor loadings by $\hat{F} = \sqrt{T} \text{eig}_K(R^\top R)$ and $\hat{B} = T^{-1} R \hat{F}$, respectively, where $\text{eig}_K(M)$ takes the eigenvectors of the matrix M corresponding to its first K largest eigenvalues.
 - (2). Compute $\hat{\mu}_f = T^{-1} \sum_{t=1}^T \hat{F}_t$, $\hat{\Sigma}_f = T^{-1} \sum_{t=1}^T \hat{F}_t \hat{F}_t^\top$, $\hat{\Sigma}_r = \hat{B}(\hat{\Sigma}_f - \hat{\mu}_f \hat{\mu}_f^\top) \hat{B}^\top + \hat{\Sigma}_{e,sp}$, where $\hat{\Sigma}_{e,sp}$ is sparse error covariance estimator defined in (3.15). Compute $\hat{V}_{g1,T}$ by (3.18), $\hat{V}_{F,T}$ and $\hat{V}_{f,i,T}$ by (3.17) with $g(x) = x$ and $g(x) = xx^\top I_i$, respectively, where $\hat{V}_{g1,T}$, $\hat{V}_{F,T}$ and $\hat{V}_{f,i,T}$ are estimators of V_{g1} , V_F and $V_{f,i}$, respectively.
 - (3). The optimal choice of δ : $1/T$ times the $(1 - \alpha_0)$ (i.e. 95%) quantile of $\hat{L}_0$, where $\hat{L}_0 = \|\hat{Z}_0\|_2^2/4 \left(1 - \hat{\mu}_f^\top \hat{\Sigma}_f^{-1} \hat{\mu}_f\right) + \sum_{i=1}^K \|\hat{\Sigma}_f^{-1/2} \hat{Z}_{2,i}\|_2^2/4$, with $\hat{Z}_0 \sim N(0, \hat{V}_{g1,T})$ and $\hat{Z}_{2,i} \sim N(0, \hat{V}_{f,i,T})$.
 - (4). Compute the estimator of w_{mv}^* , denoted by $\hat{w}_{mv}$, by replacing the population covariance matrix and population mean with $\hat{\Sigma}_r$ and $\hat{B} \hat{\mu}_f$, respectively, in the classic mean-variance problem (1.1).
 - (5). Obtain the ϵ (i.e., 5%) quantile (denoted by $\hat{A}_\epsilon$) of a normal variable whose mean is zero and variance is $\hat{w}_{mv}^\top \hat{B} \hat{V}_{F,T} \hat{B}^\top \hat{w}_{mv}$.
 - (6). The optimal choice of ρ is given by $\bar{\alpha} - \left(\sqrt{\hat{\delta}_{hd}} \|\hat{B}^\top \hat{w}_{mv}\|_2 - T^{-1/2} \hat{A}_\epsilon\right)$, where $\bar{\alpha}$ is the target portfolio return in classic mean-variance problem (1.1) and $\hat{\delta}_{hd}$ is the optimal choice of δ in Step (3).
-

Let us now proceed to detail the estimation procedure. Specifically, we begin by calculating the estimators of common factors and factor loading, which are obtained by solving

$$\min_{B \in \mathbb{R}^{p \times K}, F_t \in \mathbb{R}^K} \sum_{t=1}^T \|r_t - B F_t\|_2^2, \quad (3.13)$$

subject to the normalization for identification

$$\frac{1}{T} F F^\top = I_K, \text{ and } \frac{1}{p} B^\top B \text{ is diagonal.} \quad (3.14)$$

It was shown that $\hat{F} = \sqrt{T} \text{eig}_K(R^\top R)$ and $\hat{B} = T^{-1} R \hat{F}$ where $\text{eig}_K(M)$ takes the eigenvectors of M corresponding to its K largest eigenvalues (see, e.g., Bai and Ng, 2002; Fan et al., 2013, 2018). Then, the term $\mu_f^{*\top} \tilde{\Sigma}_f^{*-1} \mu_f^*$ can be easily estimated by its sample version $\hat{\mu}_f^\top \hat{\Sigma}_f^{-1} \hat{\mu}_f$, where $\hat{\mu}_f = T^{-1} \sum_{t=1}^T \hat{F}_t$ and $\hat{\Sigma}_f = T^{-1} \sum_{t=1}^T \hat{F}_t \hat{F}_t^\top$. It is worth noting that, under the factor model based framework, the sample counterpart $\hat{\mu}_f^\top \hat{\Sigma}_f^{-1} \hat{\mu}_f$ is consistent with its oracle version (see Theorem 3). Therefore, there is no need to employ adjusted versions as applied in, for example, Kan and Zhou (2007) and Ao et al. (2019).

Once we obtain $\hat{B}$ and $\hat{F}$, the estimated error terms $\hat{e}_t = (\hat{e}_{1t}, \dots, \hat{e}_{pt})^\top$ are naturally given by $\hat{e}_t = r_t - \hat{B} \hat{F}_t$, and therefore, the error covariance matrix can be further estimated by the adaptive sparse covariance matrix estimation $\hat{\Sigma}_{e,sp}$, of which the i th row and j th column element is given by

$$\hat{\Sigma}_{e,sp,ij} = \begin{cases} s_{ii} & i = j \\ \rho_{ij}(s_{ij}) & i \neq j \end{cases}, \quad (3.15)$$

where $s_{ij} = \frac{1}{T} \sum_{t=1}^T \hat{e}_{it} \hat{e}_{jt}$, $\rho_{ij}(\cdot)$ is shrinkage function that satisfies that for all $z \in \mathbb{R}$, $\rho_{ij}(z) = 0$ when $|z| \leq \tau_{ij}$ and $|\rho_{ij}(z) - z| \leq \tau_{ij}$, τ_{ij} is the threshold, which is set to $C \sqrt{\hat{\theta}_{ij} \omega_T}$ where C is some positive constant,¹⁰ $\hat{\theta}_{ij} = \frac{1}{T} \sum_{t=1}^T (\hat{e}_{it} \hat{e}_{jt} - s_{ij})^2$ and $\omega_T = \sqrt{1/p} + \sqrt{\log p/T}$. As a result, the estimated covariance matrix of asset return based on factor structure is given by:

$$\hat{\Sigma}_r = \hat{B}(\hat{\Sigma}_f - \hat{\mu}_f \hat{\mu}_f^\top) \hat{B}^\top + \hat{\Sigma}_{e,sp}. \quad (3.16)$$

Furthermore, with $\hat{\Sigma}_r$, $\hat{B}$, and $\hat{\mu}_f$ in hand, we can compute the estimator $\|\hat{B}^\top \hat{w}_{mv}\|_2$ where $\hat{w}_{mv}$ is the estimated optimal mean-variance portfolio from problem (1.1), with $\hat{\Sigma}_r$ and $\hat{\mu} := \hat{B} \hat{\mu}_f$ replacing the corresponding population quantities.

¹⁰ The choices of shrinkage function $\rho_{ij}(\cdot)$ and threshold parameter C are given in Sections 5.3 and 6.2, respectively.

We turn to the estimation of the long-run variance–covariance matrix V_g , which plays a vital role in the procedure of selecting both δ and ρ . For notational simplicity, let us denote $Y_{g,t} = g(F_t)$ and $\mu_g^* = E_{\mathbb{P}^*}(g(F_t))$, then asymptotic covariance matrix

$$V_g = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \sum_{t'=1}^T E_{\mathbb{P}^*} \left[(Y_{g,t} - \mu_g^*)(Y_{g,t'} - \mu_g^*)^\top \right],$$

by changing of variables, can be alternatively expressed as $V_g = \lim_{T \rightarrow \infty} \bar{V}_{g,T}$, with $\bar{V}_{g,T} = \sum_{j=-T+1}^{T-1} C_T(j)$, where

$$C_T(j) = \frac{1}{T} \sum_{t=j+1}^T E_{\mathbb{P}^*} \left[(Y_{g,t} - \mu_g^*)(Y_{g,t-j} - \mu_g^*)^\top \right] \text{ for } j \geq 0,$$

and $C_T(j) = C_T(-j)$ for $j < 0$ given F_t is a stationary process. The standard method to come up with a consistent estimator is to apply the famous heteroskedasticity and autocorrelation (HAC) kernel estimation, which is given by

$$\hat{V}_{g,T} = \hat{C}_T(0) + 2 \sum_{j=1}^{T-1} k\left(\frac{j}{q_T}\right) \hat{C}_T(j), \quad (3.17)$$

where $k(\cdot)$ is real-valued kernel function satisfying [Assumption 9](#) below, q_T is the truncation parameter,¹¹ and

$$\hat{C}_T(j) = \frac{1}{T} \sum_{t=j+1}^T \left[(\hat{Y}_{g,t} - \hat{\mu}_g)(\hat{Y}_{g,t-j} - \hat{\mu}_g)^\top \right]$$

for $j \geq 0$ where $\hat{Y}_{g,t} = g(\hat{F}_t)$ and $\hat{\mu}_g = \frac{1}{T} \sum_t \hat{Y}_{g,t}$. For ease of exposition, we denote $\hat{V}_{g,T}$ with $g(x) = x$ as $\hat{V}_{F,T}$ and with $g(x) = xx^\top \mathbf{I}_I$, $i = 1, \dots, K$, as $\hat{V}_{f,i,T}$, respectively. We note that the HAC estimators $\hat{V}_{g,T}$ is defined based on estimated factors $\hat{F}_t$ since the true latent factors F_t are unobservable in our setup. After obtaining the estimation of V_F , we can immediately estimate $w_{mv}^{*\top} B V_F B^\top w_{mv}^*$ by $\hat{w}_{mv}^\top \hat{B} \hat{V}_{F,T} \hat{B}^\top \hat{w}_{mv}$.

Finally, we still need to estimate for V_{g1} defined in [Theorem 2](#), which requires the population information of $\tilde{\Sigma}_f^{*-1} \mu_f^*$. Notice that $E_{\mathbb{P}^*}(g_1(F_t)) = 0$ where $g_1(x) = -x + xx^\top \tilde{\Sigma}_f^{*-1} \mu_f^*$. Therefore, we further define a feasible HAC estimator $\hat{V}_{g1,T}$ based on $\tilde{Y}_{g1,t} := -\hat{F}_t + \hat{F}_t \hat{F}_t^\top \tilde{\Sigma}_f^{*-1} \hat{\mu}_f$ as follows:

$$\hat{V}_{g1,T} := \tilde{C}_T(0) + 2 \sum_{j=1}^{T-1} k\left(\frac{j}{q_T}\right) \tilde{C}_T(j), \quad \tilde{C}_T(j) := \frac{1}{T} \sum_{t=j+1}^T \tilde{Y}_{g1,t} \tilde{Y}_{g1,t-j}^\top. \quad (3.18)$$

4. Theoretical guarantee of parameter estimates

We note that again, the calculations of δ and ρ from [Sections 3.1](#) and [3.2](#) require the knowledge of V_g , $\mu_f^{*\top} \tilde{\Sigma}_f^{*-1} \mu_f^*$, $w_{mv}^{*\top} B V_g B^\top w_{mv}^*$, and $\|B^\top w_{mv}^*\|_2$. In this section, we further provide the theoretical guarantee for the corresponding estimations proposed in [Section 3.3](#).

4.1. Basic assumptions

The following basic technique assumptions ensure that our estimators, developed in the previous section, are asymptotically consistent with corresponding true values.

Assumption 4. All the eigenvalues of the $K \times K$ matrix $p^{-1} B^\top B$ are bounded away from both 0 and ∞ as $p \rightarrow \infty$.

Assumption 5.

- (a) $\{e_t, F_t\}$ is strictly stationary. In addition, $E_{\mathbb{P}^*}(e_{it}) = E_{\mathbb{P}^*}(e_{it} F_{jt}) = 0$ for all $i \leq p, j \leq K$ and $t \leq T$.
- (b) There are constants $c_1, c_2 > 0$ such that $\lambda_{\min}(\Sigma_e^*) > c_1$, $\|\Sigma_e^*\|_1 < c_2$ and $\min_{i \leq p, j \leq p} \text{Var}_{\mathbb{P}^*}(e_{it} e_{jt}) > c_1$.
- (c) Let $m_\zeta = \max_{i \leq p} \sum_{j \leq p} |\Sigma_{e,ij}^*|^\zeta$ for some $\zeta \in [0, 1)$ satisfying that $\omega_T^{1-\zeta} m_\zeta = o(1)$ where $\omega_T = \sqrt{1/p} + \sqrt{\log p/T}$.
- (d) There are $y_1, y_2 > 0$ and $\beta_1, \beta_2 > 0$ such that for any $s > 0$, $i \leq p$ and $j \leq K$, $\mathbb{P}^*(|e_{it}| > s) \leq \exp\{-(s/\beta_1)^{y_1}\}$ and $\mathbb{P}^*(|F_{jt}| > s) \leq \exp\{-(s/\beta_2)^{y_2}\}$.

Assumption 6. There exists $y_3 > 0$ such that $3y_1^{-1} + 1.5y_2^{-1} + y_3^{-1} > 1$, and $C > 0$ satisfying, for all $t \in \mathbb{Z}^+$, $\alpha(t) \leq \exp(-Ct^{y_3})$, where α is mixing coefficient defined by $\sup_{A \in \mathcal{F}_{-\infty}^0, B \in \mathcal{F}_T^\infty} |\mathbb{P}^*(A) \mathbb{P}^*(B) - \mathbb{P}^*(AB)|$, $\mathcal{F}_{-\infty}^0$ and $\mathcal{F}_T^\infty$ are the σ -algebras generated by $\{(F_t, e_t) : t \leq 0\}$ and $\{(F_t, e_t) : t \geq T\}$ respectively.

¹¹ The choices of both the kernel function $k(\cdot)$ and the bandwidth parameter q_T will be specified in the subsequent simulation [Section 5.2](#).

Assumption 7. There exists constant $C > 0$ such that, for all $i \leq p$, $t \leq T$ and $s \leq T$,

- (a) $\|b_i\|_\infty < C$.
- (b) $\mathbb{E}_{\mathbb{P}^*} \left(p^{-1/2} (e_s^\top e_t - \mathbb{E}_{\mathbb{P}^*}(e_s^\top e_t)) \right)^4 < C$,
- (c) $\mathbb{E}_{\mathbb{P}^*} \|p^{-1/2} \sum_{i=1}^p b_i e_{it}\|^4 < C$.

Assumption 4 requires the factors to be pervasive, that is, to impact a non-vanishing proportion of individual asset returns. **Assumption 5(a)** requires both error term and common factors to be strictly stationary, and e_t and F_t are uncorrelated. **Assumption 5(b)** assumes that Σ_e^* is well-conditioned. **Assumption 5(c)** imposes a sparsity condition on Σ_e^* based on the intuition that many pairs of cross-sectional units become weakly correlated after removing the common factors. In particular, m_ζ is the maximum number of non-zero elements in each row when $\zeta = 0$. **Assumption 5(d)** imposes exponential-type tails for both errors and common factors. Note that it also implies that factor F_t has a finite fourth-order moment assumed in **Assumption 1(a)**. **Assumption 6** imposes a strong mixing condition on F_t and e_t , requiring the mixing coefficient α to decay exponentially. This condition further allows us to apply a kernel-based approach to consistently estimate the long-run variance–covariance matrix V_g . Without loss of generality, we assume that $\mathbb{E}_{\mathbb{P}^*}(F_t F_t^\top) = I_K$, the similar simplifications in the context of latent factor model that can be found in, for example, [Fan et al. \(2013\)](#) and [Li and Li \(2022\)](#).

Under **Assumptions 4–7**, the estimated $\hat{b}_i$ and $\hat{F}_i$ are consistent with the rotated population factor loading $H b_i$ and common factors $H F_t$, where rotation matrix $H := \frac{1}{T} \mathcal{V}^{-1} \hat{F}^\top F B^\top B$ and $\mathcal{V}$ is the $K \times K$ diagonal matrix of the first K largest eigenvalues of the sample covariance of r_t , see more details in [Fan et al. \(2013\)](#). However, the HAC estimator $\hat{V}_{g,T}$ is established on estimated factors $\hat{F}_t$, and therefore, it cannot consistently converge to the true one without additional conditions.¹² As a result, we further provide the exact identification assumption as follows.

Assumption 8. $H = I_K + O_p(\Delta)$ where $\Delta/(\sqrt{\log p/T} + T^{1/4}/\sqrt{p}) = o(1)$.

Assumption 8 requires that the rotation matrix H is approximately equal to an identity matrix. In fact, [Bai and Ng \(2013\)](#) studied conditions under which the latent factors can be estimated asymptotically without rotation. For example, if we impose the identification condition $B^\top B$ is a diagonal matrix with distinct entries, then we can derive that $H = I_K + O_p(\delta_{pT}^{-2})$ where $\delta_{pT} = \min(\sqrt{p}, \sqrt{T})$. A block diagonal matrix of factor loadings satisfies this condition. More examples can be found in [Bai and Ng \(2013\)](#).

Assumption 9.

- (a) The kernel function $k(\cdot)$ has bounded support on $[-1, 1]$ and $|k(x)| \leq 1$ for all x on the real line, $k(x) = k(-x)$, $k(0) = 1$; $k(x)$ is continuous at zero and for almost all x .
- (b) $q_T^{-1} + q_T T^{-1/2} + q_T T^{1/4} p^{-1/2} \rightarrow 0$ as $T \rightarrow \infty$.

Assumption 9(a) gives the requirement of kernel function applied in HAC estimation of V_g , which is common, see e.g., [De Jong and Davidson \(2000\)](#). Many classic commonly employed kernel functions, e.g., Bartlett kernel, quadratic spectral kernel and so on, satisfy this Assumption ([Andrews, 1991](#); [Andrews and Monahan, 1992](#)). For lag truncation parameter q_T , it is required to diverge to infinity, but the rate of divergence is also constrained by the convergence rate of $\hat{F}_t$ with respect to F_t .

Assumption 10.

- (a) Suppose $\mathbf{1}_p^\top \Sigma_r^{*-1} \mathbf{1}_p \asymp \mathbf{1}_p^\top \Sigma_r^{*-1} \mu \asymp \mu^\top \Sigma_r^{*-1} \mu \asymp p^\phi$, and $(\mathbf{1}_p^\top \Sigma_r^{*-1} \mathbf{1}_p)(\mu^\top \Sigma_r^{*-1} \mu) - (\mathbf{1}_p^\top \Sigma_r^{*-1} \mu)^2 \asymp p^{2\phi}$.
- (b) $(p^{3-3\phi} + p^{1-\phi}) \left(m_\zeta \omega_T^{1-\zeta} + T^{1/4} p^{-1/2} \right) + p^{2-2\phi} q_T (T^{-1/2} + T^{1/4} p^{-1/2}) = o(1)$.

Assumption 10(a) supposes that $\mathbf{1}_p^\top \Sigma_r^{*-1} \mathbf{1}_p$, $\mathbf{1}_p^\top \Sigma_r^{*-1} \mu$, and $\mu^\top \Sigma_r^{*-1} \mu$ have the same order of magnitude. This assumption is made for simplicity; different orders could be assumed for each term. Furthermore, the assumption that the term $\mathbf{1}_p^\top \Sigma_r^{*-1} \mathbf{1}_p$ is of the order of powers of p can also be found in [Ding et al. \(2021\)](#) and [Fan et al. \(2024b\)](#). Furthermore, **Assumption 10(a)** also indicates that the assets in the investment pool are well diversified such that μ is not close to $c_1 \mathbf{1}_p$ for some constant c_1 .

4.2. Asymptotic consistency

The following **Theorems 3** and **4** provide the consistency of $\hat{\mu}_f^\top \hat{\Sigma}_f^{-1} \hat{\mu}_f$ and the HAC estimators to their population counterparts, respectively:

Theorem 3. Suppose **Assumptions 1, 4–7** hold true, $y^{-1} = 3y_1^{-1} + 1.5y_2^{-1} + y_3^{-3} + 1$, $\log(p) = o(T^{y/6})$, and $T = o(p^2)$. Then as $T \rightarrow \infty$, we have $\hat{\mu}_f^\top \hat{\Sigma}_f^{-1} \hat{\mu}_f \xrightarrow{p} \mu_f^{*\top} \tilde{\Sigma}_f^{-1} \mu_f^*$.

¹² See more discussion in [Remark 6](#).

Remark 5. We note that all asymptotic results in this paper are derived under the regime where both p and T diverge, i.e., $p, T \rightarrow \infty$. For example, in [Theorem 3](#), the condition $T = o(p^2)$ implies that p must diverge to infinity as $T \rightarrow \infty$. Moreover, the growth rate of p is further restricted by the condition $\log(p) = o(T^{y/6})$, which indicates that p increases at a rate that is slower than an exponential function of T raised to the power of $y/6$.

Theorem 4. Under the conditions of [Theorem 3](#), and [Assumptions 8](#) and [9](#) hold true, as $T \rightarrow \infty$, we have $\hat{V}_{g,T} \xrightarrow{p} V_g$ and $\hat{V}_{g1,T} \xrightarrow{p} V_{g1}$.

Based on [Theorem 4](#), by letting $g(x) = x$ and $g(x) = xx^\top I_i$, we know that $\hat{V}_{F,T}$ and $\hat{V}_{f,i,T}$ converge to V_F and $V_{f,i}$, respectively. Furthermore, if F_t is independent over time, then V_F reduces to the covariance matrix of F_t . In this case, it is straightforward to verify that $\hat{V}_{F,T}$ simplifies to $\hat{\Sigma}_f - \hat{\mu}_f \hat{\mu}_f^\top$ by setting $q_T \leq 1$, which is a consistent estimator according to the proof of [Theorem 3](#). Based on [Theorems 3](#) and [4](#), we can consistently estimate the $1 - \alpha_0$ quantile of L_0 , and thus, we obtain the consistent estimate of δ chosen from [Section 3.1](#).

Remark 6 (Discussion Without Exact Identification Condition). The exact identification condition in [Assumption 8](#) is used to guarantee consistent estimation on population parameters in practice. Without the exact identification assumption, by applying similar arguments as those for [Theorem 4](#) under some regular conditions, it can be shown that $\hat{V}_{g,T} = \mathbb{V}_g + o_p(1)$, where $\mathbb{V}_g := \lim_{T \rightarrow \infty} \text{Var}_{\mathbb{P}^*} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T g(HF_t) \right)$. It is clear that, generally $V_g \neq \mathbb{V}_g$ unless $H = I + o_p(1)$. However, if we initially define the uncertainty quantification on the rotated factors HF_t , and impose regular assumptions on HF_t as well, we can still derive the theoretical results in all theorems by replacing F_t with HF_t . In view of this point, this exact identification condition is removable essentially but with more tedious assumptions and proofs.

Back to the asymptotic L_0 in [Theorem 2](#), without [Assumption 8](#), only the estimation of the second term in L_0 will be affected while the first term of L_0 is not affected in estimation. In details, by some algebra, it follows that $\hat{V}_{g1,T} = HV_{g1}H^\top + o_p(1)$. Notice that the estimation $\hat{V}_{g1,T}$ is obtained to deal with the unknown normal variate $Z_0 \sim N(0, V_{g1})$ in [Theorem 2](#). With $\hat{V}_{g1,T}$ in hand, we can use $\hat{Z}_0 \sim N(0, \hat{V}_{g1,T})$ to replace Z_0 . Rewrite $Z_0 = V_{g1}^{1/2} Z_0^*$, $Z_0^* \sim N(0, I_K)$, then $\hat{Z}_0$ can be expressed as $HV_{g1}^{1/2} Z_0^* + o_p(1)$. It is clear that $\|Z_0\|_2^2 = Z_0^{*\top} V_{g1} Z_0^*$ can be approximated by $\|\hat{Z}_0\|_2^2 = Z_0^{*\top} V_{g1}^{1/2} H^\top H V_{g1}^{1/2} Z_0^* + o_p(1) = \|Z_0\|_2^2 + o_p(1)$ given that $H^\top H = I_K + o_p(1)$ ([Lemma 3](#) in the supplementary material) even if [Assumption 8](#) violates. As a result, the component of the limit L_0 involving Z_0 is not affected by [Assumption 8](#).

In addition, in [Section B.4](#) of the supplementary material, we also provide some simulation evidence, which shows that the estimated optimal uncertainty size remains close to the true level even when [Assumption 8](#) violates. In more detail, we set the factor loading matrix B such that $B^\top B$ is a diagonal matrix with identical entries. From [Table B.8](#) in the supplementary material, it can be observed that the mean and median of the estimated uncertainty size are close to the true levels, with the ratios (estimated value over corresponding true value) in brackets being at least 0.9.

Next, the following [Theorem 5](#) provides the consistency of estimated variance of normal variate Z_1 in [\(3.11\)](#).

Theorem 5. Under the conditions of [Theorem 3](#), and [Assumptions 9](#) and [10](#) hold true, then we have $\hat{w}_{mv}^\top \hat{B} \hat{V}_{F,T} \hat{B}^\top \hat{w}_{mv} \xrightarrow{p} w_{mv}^{*\top} B V_F B^\top w_{mv}^*$.

As an immediate result of [Theorem 5](#), it is easy to know that $\|\hat{B}^\top \hat{w}_{mv}\|_2$ is consistent with $\|B^\top w_{mv}^*\|_2$ by letting $\hat{V}_{F,T} = V_F = I_K$. Consequently, based on [Theorem 5](#), the optimal value ρ_{hd} selected from [Section 3.2](#) can be consistently estimated.

5. Simulation

In this section, we conduct Monte Carlo simulations to evaluate the performance of the selection procedure of δ and ρ as well as the newly proposed high-dimensional distributionally robust mean–variance portfolio.

5.1. Basic setup

The virtual return data are generated by a factor model defined in [\(2.4\)](#) with two common factors. We impose an AR(1) structure to each factor, that is

$$f_{i,t} = \beta_i + \alpha_i f_{i,t-1} + v_{i,t},$$

where $v_i \sim N(0, (1 - \alpha_i^2)(1 - \beta_i^2)/(1 - \alpha_i)^2)$ such that $E_{\mathbb{P}^*}(f_{i,t}^2) = 1$. The AR(1) specification for latent common factors is often applied in factor model-based financial literature; see e.g., [Su and Wang \(2017\)](#) and [Fan et al. \(2024b\)](#). For factor loading B , we set it to a block diagonal matrix

$$B = \begin{pmatrix} B_1 & \mathbf{0}_{p_1} \\ \mathbf{0}_{p_2} & B_2 \end{pmatrix},$$

where B_i is a p_i -dimensional column vector with $p_1 + p_2 = p$, and $B_i \sim N(\mu_{b_i}, \sigma_{b_i}^2)$. For error term e_t , we simply draw them from multivariate normal distributions $N(0, \Sigma_e)$. The values of population parameters are calibrated from real monthly return data of the

components of the S & P 500 index from 01/1990 to 12/1999.¹³ In more detail, we apply classic principal component analysis to estimate the latent factors, factor loadings, and corresponding residuals from real data. Parameters α_i and β_i are set to the autocoefficient and intercept estimations fitted by the AR(1) process using estimated factors, and μ_{b_i} and $\sigma_{b_i}^2$ are given by the mean and variance of estimated loading. Thus $\beta_1 = 0.4313, \beta_2 = -0.0036, \alpha_1 = 0.0305, \alpha_2 = 0.1032, \mu_{b_1} = 0.0494, \mu_{b_2} = -0.0100, \sigma_{b_1} = 0.0227, \sigma_{b_2} = 0.0232$. For Σ_e^* , we assign the estimated sparse error covariance matrix (3.15) based on estimated residuals with threshold $\tau_{ij} = 0.5\sqrt{\hat{\theta}_{ij}\omega_T}$ where $\hat{\theta} = \frac{1}{T} \sum_{i=1}^T (\hat{e}_{it}\hat{e}_{jt} - s_{ij})^2$. In this section, we assume that the number of factors is known to eliminate the impact of incorrect selection of common factors. In Section B.1 of our supplementary material, we also report the results when the factor number K is estimated, the results only show minor discrepancies.

5.2. Performance of selected δ

In this subsection, we evaluate the estimated value of δ selected from Section 3.1 via simulation. Since all population parameters are calibrated based on monthly return data, we set the target portfolio return $\bar{\alpha}$ to 2%, which corresponds to an annualized excess return of 26.82%. We highlight that only the selection procedure used by BCZ requires the input of target portfolio return, our method is designed to be free of $\bar{\alpha}$ due to the minimum variance problem applied. We generate 500 virtual dataset $R_{p \times T}$ with $T = 250$ and $p \in \{30, 50, 80, 100\}$. For HAC estimators $\hat{V}_{g1,T}$ and $\hat{V}_{f,i,T}$, we choose Bartlett kernel function $k(x) = (1 - |x|)I(x \leq 1)$ which can be verified to satisfy Assumption 9, see Andrews (1991), and bandwidth $q_T = \lfloor 5T^{-1/8}p^{1/4} \rfloor$ given that $q_T T^{1/4} p^{-1/2} = o(1)$. For notational simplicity, the selected δ at population level from BCZ and our method are denoted as δ_{bl} and δ_{hd} , with corresponding estimated versions denoted as $\hat{\delta}_{bl}$ and $\hat{\delta}_{hd}$, respectively.

Table 1 reports the median and mean level of $\hat{\delta}_{bl}$ and $\hat{\delta}_{hd}$ over 500 replications as well as the corresponding oracle values. The values in brackets are the ratios of estimated values over the corresponding population values, indicating the deviation from the true level. First, as we mentioned before, it is evident that the median and mean level of $\hat{\delta}_{bl}$ largely deviate from the true δ_{bl} . In contrast, those of $\hat{\delta}_{hd}$ are close to the corresponding true values in all cases. For example, in the case of $p = 100$, the mean of $\hat{\delta}_{bl}$ at the 90% confidence level is 14.39E-03, which is about 3.5 times the oracle value, whereas the value of $\hat{\delta}_{hd}$ is approximately equal to the corresponding oracle value. In our additional simulation results presented in Section B.3 of the supplementary material, the mean of $\hat{\delta}_{bl}$ can exceed the corresponding true level by more than 2500 times when the population parameters are calibrated from daily data. Furthermore, from Table 1, it can be observed that the deviation of $\hat{\delta}_{bl}$ from the true level increases substantially as p gets larger. We note that the comparison based on absolute values between BCZ method and our method is not meaningful since the values of F_t and r_t are not directly comparable due to the existence of factor loading.

To see more insights into estimated values of δ , Table 2 further presents the standard deviation and the maximum value of both $\hat{\delta}_{bl}$ and $\hat{\delta}_{hd}$ over 500 simulation replications. The values in brackets are the ratios compared to the corresponding median levels. First, it can be observed that the standard deviation ratios and the maximum value ratios of $\hat{\delta}_{bl}$ are nearly twice and 1.3 times those of $\hat{\delta}_{hd}$, respectively. For example, in the case of $p = 80$ at 95% confidence level, the standard deviation ratio and maximum value ratio of $\hat{\delta}_{bl}$ are 0.19 and 1.70, which are about 2.1 and 1.3 times those of $\hat{\delta}_{hd}$. It indicates that $\hat{\delta}_{bl}$ is a more unstable estimator compared to $\hat{\delta}_{hd}$, which is detrimental to a method known for its robustness, and hence our proposal is favored in practical application. It is also worth mentioning that, in Section B.3 of the supplementary material, the standard deviations and maximum values of $\hat{\delta}_{bl}$ can exceed the median level by more than 1500 and 45,000 times, respectively, which is quite alarming. In contrast, the fluctuations of our $\hat{\delta}_{hd}$ are much more stable, with the standard deviations and maximum values maintained at approximately 0.1 and 1.3 times the corresponding median level, respectively.

For a distributionally robust model, it is interesting to study the relationship between model uncertainty and the selected δ_{hd} . It is expected that the value of δ_{hd} increases as model uncertainty increases. For the factors with AR(1) structure, an easy way to alter the uncertainty is to change the autocoefficient α_i . When α_i is close to zero, the factor tends to be independent across the time horizon. When α_i is close to 1, the factor has a long persistent correlation. Clearly, long-term correlation will result in more uncertainty than independence. Let $\alpha_i = J\alpha_{i0}$ where α_{i0} is the value calibrated from real data, and the previous results are based on the situation of $J = 1$. Fig. 1 plots uncertainty measures δ_{hd} with respect to the cases of different $J \in \{1, 2, \dots, 6\}$ at the 0.9, 0.95 and 0.99 confidence levels. It is evident that δ_{hd} takes larger value as J gets larger, indicating that the model uncertainty increases as J gets larger. Thus, we can conclude that the value of δ from our selection procedure well reflects the size of model uncertainty.

5.3. Performance of selected ρ

In this subsection, we further study the estimated value of ρ from Section 3.2. First, recall that the choice of ρ is given by $\bar{\alpha} - \sqrt{\delta} \|B^T w_{mv}^*\|_2 + A_\epsilon / \sqrt{T}$ where A_ϵ is the ϵ quantile of a normal variate Z_1 in (3.11). In this subsection, the target return $\bar{\alpha}$ is still set to 2%. For uncertainty size δ , the estimated uncertainty size at the 95% confidence level is used. As such, our results actually reflect the quality of the selected value of ρ when investors follow our suggested procedure in practical applications. Since the variance of the normal variate Z_1 depends on the optimal mean–variance allocation w_{mv}^* , the estimation of return covariance matrix is also required. For shrinkage function $\rho_{ij}(\cdot)$, the soft thresholding function is applied with $C = 0.5$ following the studies

¹³ The historical data at the first decision node in our empirical analysis based on monthly data.

Table 1

The median and mean level of $\hat{\delta}_{bl}$ and $\hat{\delta}_{hd}$, and corresponding oracle values. The values in brackets are the ratios compared to the corresponding population value. The target return $\bar{\alpha}$ is set to 2%. The 90%, 95% and 99% confidence levels for choosing δ are considered.

p	$\hat{\delta}_{bl}(\times 10^{-3})$			$\hat{\delta}_{hd}(\times 10^{-3})$		
	90%	95%	99%	90%	95%	99%
Oracle						
30	1.34	1.53	1.95	14.51	17.46	24.12
50	2.19	2.47	3.13	14.47	17.41	24.00
80	3.27	3.69	4.72	14.47	17.38	24.13
100	4.08	4.62	6.00	14.49	17.41	24.03
Median						
30	2.15 (1.60)	2.44 (1.60)	3.16 (1.62)	13.86 (0.96)	16.71 (0.96)	23.32 (0.97)
50	4.36 (1.99)	4.95 (2.01)	6.32 (2.02)	13.77 (0.95)	16.60 (0.95)	23.14 (0.96)
80	9.08 (2.78)	10.27 (2.79)	13.16 (2.79)	13.82 (0.96)	16.64 (0.96)	23.06 (0.96)
100	14.24 (3.49)	16.10 (3.49)	20.70 (3.45)	13.91 (0.96)	16.75 (0.96)	23.17 (0.96)
Mean						
30	2.14 (1.60)	2.46 (1.61)	3.21 (1.64)	13.95 (0.96)	16.84 (0.96)	23.47 (0.97)
50	4.43 (2.02)	5.05 (2.04)	6.45 (2.06)	13.86 (0.96)	16.77 (0.96)	23.35 (0.97)
80	9.24 (2.83)	10.48 (2.84)	13.37 (2.83)	13.94 (0.96)	16.81 (0.97)	23.45 (0.97)
100	14.39 (3.53)	16.36 (3.54)	21.03 (3.51)	14.01 (0.97)	16.90 (0.97)	23.58 (0.98)

Table 2

The standard deviation and the maximum value of $\hat{\delta}_{bl}$ and $\hat{\delta}_{hd}$ over simulation replications. The values in bracket are the ratio compared to the corresponding median level. The target return $\bar{\alpha}$ is set to 2%. The 90%, 95% and 99% confidence levels for choosing δ are considered.

p		$\hat{\delta}_{bl}(\times 10^{-3})$			$\hat{\delta}_{hd}(\times 10^{-3})$		
		90%	95%	99%	90%	95%	99%
30	SD	0.44 (0.21)	0.51 (0.21)	0.70 (0.22)	1.18 (0.09)	1.53 (0.09)	2.58 (0.11)
		3.52 (1.64)	4.05 (1.66)	5.35 (1.70)	18.63 (1.34)	21.95 (1.31)	32.00 (1.37)
	Max	0.81 (0.19)	0.94 (0.19)	1.26 (0.20)	1.08 (0.08)	1.36 (0.08)	2.30 (0.10)
50	SD	7.36 (1.69)	8.43 (1.70)	10.58 (1.67)	17.97 (1.30)	21.53 (1.30)	32.10 (1.39)
		15.82 (1.74)	17.46 (1.70)	23.18 (1.76)	17.65 (1.28)	22.36 (1.34)	31.66 (1.37)
	Max	1.64 (0.18)	1.90 (0.19)	2.58 (0.20)	1.11 (0.08)	1.45 (0.09)	2.55 (0.11)
80	SD	2.48 (0.17)	2.87 (0.18)	3.99 (0.19)	1.16 (0.08)	1.52 (0.09)	2.55 (0.11)
		22.99 (1.61)	27.02 (1.68)	39.30 (1.90)	19.63 (1.41)	24.70 (1.47)	36.70 (1.58)
	Max	1.64 (0.18)	1.90 (0.19)	2.58 (0.20)	1.11 (0.08)	1.45 (0.09)	2.55 (0.11)
100	SD	2.48 (0.17)	2.87 (0.18)	3.99 (0.19)	1.16 (0.08)	1.52 (0.09)	2.55 (0.11)
		22.99 (1.61)	27.02 (1.68)	39.30 (1.90)	19.63 (1.41)	24.70 (1.47)	36.70 (1.58)
	Max	1.64 (0.18)	1.90 (0.19)	2.58 (0.20)	1.11 (0.08)	1.45 (0.09)	2.55 (0.11)

of Fan et al. (2013, 2024b). The other simulation setups remain unchanged from those used in the previous subsection. We denote the estimated (oracle) choices from our work and those from BCZ by $\hat{\rho}_{hd}$ (ρ_{hd}) and $\hat{\rho}_{bl}$ (ρ_{bl}), respectively.

Table 3 reports the median and mean level of $\hat{\rho}_{hd}$ and $\hat{\rho}_{bl}$ over 500 replications as well as the corresponding oracle values. We still consider 90%, 95%, and 99% confidence levels. The values in brackets are the differences between estimated value to the corresponding true level. First of all, it is evident that $\hat{\rho}_{bl}$ severely deviates from the corresponding true level in all cases, while $\hat{\rho}_{hd}$ is quite close to the oracle level. For example, in the case of $p = 100$, across all three confidence levels, the biases between the mean of $\hat{\rho}_{bl}$ and the corresponding true value are at least 4%, whereas those for $\hat{\rho}_{hd}$ are at most 0.05%. Furthermore, the deviation

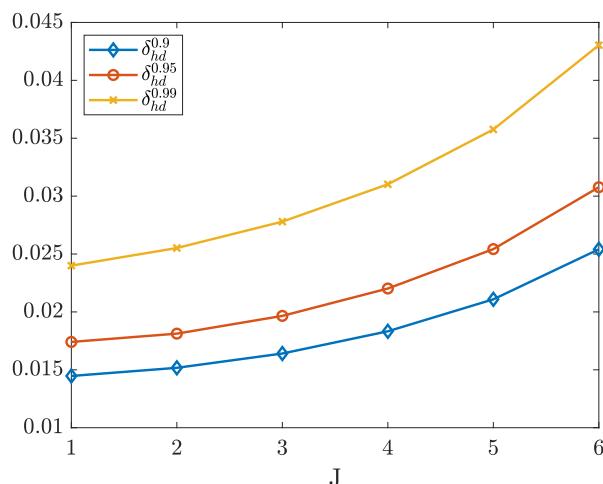


Fig. 1. The model uncertainty measured by δ_{hd} under confidence levels 0.9, 0.95, and 0.99.

Table 3

The median and mean level of $\hat{\rho}_{bl}$ and $\hat{\rho}_{hd}$, and corresponding oracle values. The values in brackets are the differences between mean/median level and corresponding oracle level. The target return $\bar{\alpha}$ is set to 2%. The 1%, 5% and 10% significance levels are considered. All values in the table are expressed as percentages (%).

p	$\hat{\rho}_{bl}$			$\hat{\rho}_{hd}$		
	1%	5%	10%	1%	5%	10%
Oracle						
30	0.158	0.370	0.482	0.724	0.924	1.030
50	0.241	0.447	0.557	0.725	0.925	1.031
80	0.270	0.474	0.583	0.726	0.925	1.032
100	0.269	0.472	0.580	0.725	0.925	1.031
Median						
30	-0.510 (-0.668)	-0.648 (-1.018)	-0.932 (-1.415)	0.719 (-0.005)	0.912 (-0.012)	1.016 (-0.015)
50	-1.275 (-1.516)	-1.450 (-1.897)	-1.824 (-2.382)	0.765 (0.040)	0.951 (0.026)	1.049 (0.017)
80	-2.839 (-3.110)	-3.113 (-3.587)	-3.685 (-4.269)	0.791 (0.065)	0.967 (0.042)	1.065 (0.033)
100	-4.184 (-4.452)	-4.554 (-5.026)	-5.318 (-5.898)	0.801 (0.076)	0.980 (0.055)	1.075 (0.043)
Mean						
30	-0.529 (-0.687)	-0.666 (-1.035)	-0.954 (-1.436)	0.678 (-0.046)	0.876 (-0.048)	0.981 (-0.049)
50	-1.274 (-1.515)	-1.453 (-1.900)	-1.825 (-2.382)	0.737 (0.012)	0.926 (0.001)	1.026 (-0.005)
80	-2.853 (-3.124)	-3.133 (-3.607)	-3.721 (-4.305)	0.763 (0.037)	0.947 (0.021)	1.045 (0.013)
100	-4.215 (-4.483)	-4.588 (-5.059)	-5.390 (-5.970)	0.775 (0.050)	0.957 (0.032)	1.054 (0.023)

of $\hat{\rho}_{bl}$ from the true level increases sharply with the growth of p . The simulation results suggest that, investors can obtain values of $\hat{\rho}_{hd}$ that are quite close to the true level ρ_{hd} by following our selection procedure, whereas the values of ρ selected using BCZ's procedure deviate substantially from their optimal counterparts.

In addition, it can be observed that the selected values of ρ are all less than the given target return $\bar{\alpha}$, reflecting the impact of incorporating uncertainty into the portfolio decision. As we mentioned before, ρ is the lowest acceptable target return and is set for the portfolio in the worst scenario when taking into account the distributional uncertainty. Therefore, ρ is logically expected to be smaller than $\bar{\alpha}$, as confirmed by our simulation results. It is also noted that the $\hat{\rho}_{bl}$ is severely underestimated in most simulation replications, which makes the target return constraint relaxed. As a consequence, the classic BCZ optimization with selected ρ and δ does not actually perform a mean-variance tradeoff. This also explains why the BCZ model approximates an equal-weighting

strategy when $\delta \rightarrow \infty$, which corresponds to the solution of the global minimum variance problem with the covariance matrix being the identity matrix. Lastly, comparing oracle ρ_{hd} with oracle ρ_{bl} , we observe that the distributionally-robust portfolio of BCZ requires a much lower target return than our high-dimensional distributionally robust portfolio.

5.4. Performance of portfolio

In the previous two subsections, we have demonstrated that the estimating procedure in Algorithm 1 can obtain accurate and stable $\hat{\delta}_{hd}$ and $\hat{\rho}_{hd}$. In this subsection, we further study the performance of the proposed high-dimensional distributionally robust portfolio (HD-DRO) obtained from solving problem (2.7).

In order to conduct an out-of-sample performance evaluation, we generate 500 virtual datasets $R_{p \times (T_0+T)} = [R_{1,p \times T_0}, R_{2,p \times T}]$ with $T_0 = 250$ and $T = 150$. The first 250 periods of data $R_{1,p \times T_0}$ are used as the training set to construct various portfolios, and the last 150 periods of data $R_{2,p \times T}$ are treated as the testing set. Besides the results of our estimated HD-DRO, we also consider (1) true high-dimensional distributionally robust mean-variance portfolio (HD-DRO-pop) using δ_{hd} and ρ_{hd} , (2) classic distributionally robust mean-variance portfolio from BCZ (DRMV) using $\hat{\delta}_{bl}$ and $\hat{\rho}_{bl}$, (3) true distributionally robust mean-variance portfolio (DRMV-pop) using δ_{bl} and ρ_{bl} , (4) population mean-variance portfolio (MV-pop), (5) global minimum variance portfolio (GMVP- ∞) with covariance being $\delta BB^T + \Sigma_e^*$ and $\delta = 10^3$, and (6) the equal weighting strategy ($1/N$). The portfolios GMVP- ∞ and $1/N$ are special cases of HD-DRO-pop and DRMV-pop, respectively, when δ tends to infinity and the target return constraints are slack. Note that MV-pop, HD-DRO-pop, DRMV-pop and GMVP- ∞ are infeasible portfolios in practical application because they all require the population parameter information. The confidence level for choosing δ and ρ is set to 95%. For the evaluation measures, we report the out-of-sample mean (Mean), out-of-sample standard deviation (SD), and out-of-sample Sharpe ratio (defined as the ratio of the mean to the standard deviation, SR) of realized out-of-sample portfolio excess returns for all the strategies considered. Additionally, for two high-dimensional distributionally robust portfolios and two classic distributionally robust portfolios of BCZ, we also present the target optimal returns (TOR), defined as $\rho + \sqrt{\delta} \|B^T w\|_2$ and $\rho + \sqrt{\delta} \|w\|_2$, respectively. Clearly, the TOR for MV-pop is $\bar{\alpha}$.

Table 4 reports the mean levels of four measures over 500 simulation replications for the cases of $p \in \{30, 50, 80, 100\}$. First, it can be observed that the HD-DRO achieves the highest Sharpe ratio among all feasible portfolios, and its performance is quite close to the corresponding oracle version HD-DRO-pop. For example, in the case of $p = 50$, the out-of-sample SR of HD-DRO is 0.4615, which is very close to the true level of HD-DRO-pop (0.4685) and substantially outperforming DRMV, whose SR is only 0.1920. Second, the DRMV obtains a low portfolio risk, which is expected, as the $\hat{\rho}_{bl}$ is severely underestimated, leading to a relaxed target return constraint in many simulation replications. As a result, it degenerates into a penalized global minimum variance portfolio, and achieves the lowest risk among feasible portfolios. Due to the poorly selected δ and ρ , it is evident that DRMV significantly deviates from its true counterpart DRMV-pop, which also enjoys an acceptable portfolio SR in most cases. Additionally, it is worth noting that, although the overestimated $\hat{\delta}_{bl}$ does not fully push DRMV into the $1/N$ allocation in terms of reported mean values, there are still cases in our simulation experiments where the DRMV is very close to equal weighting strategy. Third, the optimal SR and risk are obtained by infeasible MV-pop and GMVP- ∞ , respectively, where the former is the theoretical optimal (in terms of SR) mean-variance portfolio given $\bar{\alpha}$, and the latter is a global minimum variance portfolio. Lastly, from the columns of target optimal return (TOR), we observe that the target portfolio return at the optimal allocation for HD-DRO-pop and DRMV-pop are approximately 1.3% and 1.1% respectively, given $\bar{\alpha} = 2\%$. This suggests that the distributionally robust portfolio is achieved by imposing a weaker target return constraint, which reduces risk to enhance robustness, while only slightly sacrificing SR. It seems that DRMV-pop is a more conservative allocation than HD-DRO-pop since its TOR is lower. Interestingly, in terms of the average level over four scenarios, a risk reduction of 0.0136 leads to a sacrifice of only 0.0131 in SR for HD-DRO-pop, whereas a larger risk reduction of about 0.0162 results in a much greater SR loss of 0.0326 for DRMV-pop, suggesting that HD-DRO-pop is a more efficient allocation strategy.

In the Section B.3 of the supplementary material, we report additional simulation results for considered portfolios when population parameters are calibrated from real daily return data. The simulation results further demonstrate the superiority of the proposed HD-DRO and are thus omitted here for brevity. However, it is worth mentioning that, with a target return $\bar{\alpha} = 0.05\%$, the selected optimal ρ and the target optimal return for all four distributionally robust portfolios are often negative, leading their behavior to closely resemble that of the corresponding global minimum variance portfolio. The negative values of ρ are primarily attributed to the low target return $\bar{\alpha} = 0.05\%$ applied. A negative ρ implies that investors require the allocation to prevent losses exceeding $-\rho$ even in the worst scenario, which fully reflects the robustness of portfolio decisions under uncertainty. In these cases, the absolute value of negative ρ can be interpreted as the maximum acceptable loss.

6. Empirical analysis

In this section, we evaluate the out-of-sample performance of our proposed high-dimensional distributionally robust portfolio using real market data. Specifically, we construct portfolios based on a rolling-window scheme. At each decision point t_c , we use the historical excess return data of length T (before t_c) to construct portfolios, which are then held for the next T_h periods to gain the realized excess return. After holding T_h periods, we re-calculate and re-balance the portfolio weights using historical data at the time point $t_c + T_h + 1$. This process is repeated until the end of the out-of-sample period.

Table 4

The out-of-sample mean, out-of-sample standard deviation, out-of-sample Sharpe ratio and target optimal return of all considered portfolios. The values of δ and ρ are chosen at the confidence level 0.95, and the target return $\bar{\alpha}$ is set to 2%. “–” means no results.

	Mean	SD	SR	TOR	Mean	SD	SR	TOR
	p = 30				p = 50			
MV-pop	0.0202	0.0434	0.4677	0.0200	0.0202	0.0420	0.4831	0.0200
HD-DRO	0.0132	0.0303	0.4373	0.0128	0.0136	0.0296	0.4615	0.0133
HD-DRO-pop	0.0133	0.0297	0.4499	0.0131	0.0133	0.0284	0.4685	0.0131
DRMV	0.0049	0.0195	0.2507	0.0044	0.0027	0.0141	0.1920	–0.0019
DRMV-pop	0.0111	0.0270	0.4138	0.0110	0.0117	0.0260	0.4514	0.0116
GMVP- ∞	0.0000	0.0182	0.0004	–	0.0000	0.0133	–0.0036	–
1/N	0.0114	0.0284	0.4035	–	0.0114	0.0259	0.4405	–
	p = 80				p = 100			
MV-pop	0.0202	0.0414	0.4889	0.0200	0.0202	0.0412	0.4921	0.0200
HD-DRO	0.0136	0.0290	0.4722	0.0135	0.0137	0.0288	0.4777	0.0136
HD-DRO-pop	0.0132	0.0278	0.4782	0.0131	0.0133	0.0275	0.4829	0.0131
DRMV	0.0020	0.0113	0.1786	–0.0168	0.0021	0.0109	0.1949	–0.0292
DRMV-pop	0.0117	0.0251	0.4659	0.0116	0.0117	0.0250	0.4694	0.0116
GMVP- ∞	–0.0001	0.0101	–0.0084	–	0.0000	0.0093	–0.0053	–
1/N	0.0113	0.0247	0.4596	–	0.0117	0.0254	0.4601	–

6.1. Choice of common factor number

In practical implementation, the number of common factors is unknown and needs to be estimated, thus we apply the standard and commonly used approach proposed by Bai and Ng (2002) to determine the number of common factors K . In detail, we solve the problem as follows:

$$\hat{K} = \arg \min_{0 \leq K \leq M} \log \left(\frac{1}{pT} \left\| R - T^{-1} R \hat{F}(K) \hat{F}(K)^\top \right\|_F^2 \right) + Kg(T, p) \quad (6.1)$$

where M is a prescribed upper bound for the number of factors,¹⁴ $\hat{F}(K)$ is estimated common factor conditional on factor number K , and $g(T, p_m)$ is a penalty function of p and T , for example, $g(T, p) = \frac{p+T}{pT} \log(\frac{pT}{p+T})$. The conditions assumed in Bai and Ng (2002) can be guaranteed by our Assumptions. As a result, the consistency of $\hat{K}$ to K is ensured by their Theorem 2.

6.2. Choice of threshold parameter

Recall that the thresholding parameter is defined as $\tau_{ij} = C\sqrt{\hat{\theta}_{ij}\omega_T}$. In implementation, we also need to choose the value for C . In more detail, we use the data-driven m -fold cross-validation method to determine it (Bickel and Levina, 2008; Fan et al., 2013, 2024b). After we obtain the estimation of error term $\hat{e}_t$, we randomly divide the residual sequence into two subsets, denoted as $\{\hat{e}_t\}_{t \in A}$ and $\{\hat{e}_t\}_{t \in B}$ respectively. Subset A is used for training, and subset B is used for testing purposes. Then, we can choose the value of C that minimizes the following objective function over a compact interval:

$$C^* = \arg \min_{\underline{C} < C_1 \leq \bar{C}} \frac{1}{m} \sum_{j=1}^m \left\| \hat{\Sigma}_e^{A,j}(C_1) - S_e^{B,j} \right\|_F^2,$$

where $\underline{C}$ is the minimum constant that guarantees the positive definiteness of $\hat{\Sigma}_e^{A,j}(C_1)$, and $\bar{C}$ is large constant such that $\hat{\Sigma}_e^{A,j}(C_1)$ is diagonal, $\hat{\Sigma}_e^{A,j}(C_1)$ is the threshold residual covariance estimator by using subset A in the j th loop with threshold C_1 , and $S_e^{B,j}$ is sample covariance matrix by using subset B in the j th loop. We set the training set to include 2/3 of the total observations.

6.3. Measures and benchmarks

The performance of the portfolio is primarily evaluated by its out-of-sample cumulative excess return (CR), risk (Risk), Sharpe ratio (SR), and maximum drawdown (MDD), which are defined as follows respectively:

$$\begin{aligned} \text{CR} &= \sum_{t=1}^T r_t^p, \quad \text{Risk} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T \left(r_t^p - \frac{\text{CR}}{T} \right)^2}, \\ \text{SR} &= \frac{\text{CR}}{T \times \text{Risk}}, \quad \text{MDD} = \max_{1 \leq t_1 \leq t_2 \leq T} - \sum_{t=t_1}^{t_2} r_t^p, \end{aligned}$$

¹⁴ In this paper, we set M to 10.

where T is the number of out-of-sample observations and r_t^p is out-of-sample portfolio excess return. In addition, as suggested by one of the reviewers, to provide a sense of the practical feasibility of the portfolios, we also report the results for portfolio turnover (TO), the amount of negative weights (NS), maximum weights (MAX), and minimum weights (MIN), defined as follows, respectively:

$$\begin{aligned} \text{TO} &= \frac{1}{RT} \sum_{l=1}^{RT} \sum_{i=1}^p |\widehat{W}_{l+1,i} - \widehat{W}_{l,i}^+|, \quad \text{NS} = \frac{1}{RT} \sum_{l=1}^{RT} \sum_{i=1}^p |\widehat{W}_{l,i}| \mathbf{I}_{\{\widehat{W}_{l,i} < 0\}}, \\ \text{MAX} &= \max_i \frac{1}{RT} \sum_{l=1}^{RT} \widehat{W}_{l,i}, \quad \text{MIN} = \min_i \frac{1}{RT} \sum_{l=1}^{RT} \widehat{W}_{l,i}, \end{aligned}$$

where $\widehat{W}_{l+1}$ is the estimated portfolio weight at $(l+1)$ th rebalancing, and $\widehat{W}_l^+$ represents the portfolio weight before the $(l+1)$ th rebalancing,¹⁵ RT is the number of rebalancing, and $\mathbf{I}_{\{\cdot\}}$ is the indicator function.

In practical applications, transaction costs play a crucial role for investors. Following DeMiguel et al. (2009b), Ao et al. (2019) and Fan et al. (2024b), the excess portfolio return net of transaction cost is computed as,

$$r_l^{\text{net}} = \left(1 - \sum_{i=1}^p c |\widehat{W}_{l+1,i} - \widehat{W}_{l,i}^+| \right) (1 + r_l) - 1, \quad (6.2)$$

where r_l is the excess return of the portfolio without transaction cost at the time of the l th rebalancing. The parameter c controls the level of the transaction cost, and in this study, it is set to 10 basis points following Ao et al. (2019), Ke et al. (2019) and Li et al. (2022).

For benchmark strategies, we consider the following 12 portfolios: (1) DRO: the distributionally robust mean–variance portfolio of BCZ with uncertainty size and target return selected from the menu at the end of their Section 4 with 95% confidence level. (2) 1/N: the equal weighting strategy (DeMiguel et al., 2009b). (3) MV: the classic mean–variance portfolio based on sample mean and sample covariance. (4) MV-L1: the regularized mean–variance portfolio with l_1 norm (DeMiguel et al., 2009a), the regularization parameter is chosen through cross-validation with an out-of-sample risk targeting loss function. (5) MV-POET: the mean–variance portfolio based on the POET covariance proposed by Fan et al. (2013) and factor model-based mean estimators $\widehat{B}\widehat{\mu}_f$, the threshold parameter is chosen by the procedure introduced in Section 6.2. (6) MV-POETS: the mean–variance portfolio based on the POET covariance matrix and sample mean estimators, the threshold parameter is chosen by the procedure introduced in Section 6.2. (7) MAXSER: the mean–variance portfolio developed by Ao et al. (2019) with sparse tuning parameter selected via their ten-fold cross-validation and using 30 assets in the sub-pool selection, the factor-investing version of MAXSER is applied with Fama–French three factors.¹⁶ (8) SRM: the mean–variance portfolio based on synthetic regression proposed by Li et al. (2022) with Fama–French three factors and tuning parameter λ selected by their ten-fold cross-validation. (9) CP-LW: the optimal combining portfolio developed by Kan et al. (2022) with linear shrinkage covariance matrix of Ledoit and Wolf (2004), the risk aversion parameter γ is set to 3. (10) CP-SF: the optimal combining portfolio developed by Kan et al., 2022 with MacKinlay–Pástor single factor structure, the semianalytical solution in Tu and Zhou (2011) is applied and the risk aversion parameter γ is set to 3. (11) R-MVP: the robust minimum variance portfolio developed by Fan et al. (2024a) with the threshold of Huber loss function set to the 0.9 quantile of the spectral norm of the estimated residuals and the threshold parameter for sparse error covariance matrix is determined by their cross-validation. (12) RSP: the robust shrinkage portfolio that exploits the intensity maximizing the mean–risk out-of-sample utility criterion proposed by Lassance et al. (2024a) with a fixed value of $\lambda = 2$ in their paper, and the risk aversion coefficient γ is set to 3. (13) BonaFide: the mean–variance shrinkage portfolio developed by Bodnar et al. (2022) with equally weighted target and risk aversion parameter $\gamma = 3$ in their study.

The results of DRO serve as the main benchmarks. Note that, the empirical analysis of BCZ indicates that the DRO demonstrates its superiority over some robust models, such as Goldfarb and Iyengar (2003). For better comparison and simplicity, we do not focus on the models that have already been considered by BCZ. Strategies MV, MV-L1, MV-POET, MV-POETS, MAXSER, and SRM are (penalized) mean–variance portfolios. Strategy R-MVP is the global minimum variance portfolio. Strategies CP-LW, CP-SF, RSP, and BonaFide are shrinkage portfolios. Our strategy is simply labeled as HD-DRO. The values of uncertainty size δ and target return ρ are selected by our Algorithm 1 with confidence level 95%, the factor number K is estimated by (6.1), and the threshold parameter C is chosen by the way in Section 6.2.

6.4. Data

We conduct our empirical analysis based on monthly return data of component stocks of the S & P 500 index and Russell 2000 index, sourced from the Center for Research in Security Prices (CRSP).¹⁷ The S & P 500 index consists of large-cap stocks, representing the largest publicly traded companies in the U.S., while the Russell 2000 index comprises small-cap stocks, focusing on smaller, growth-oriented companies. We delete those stocks that do not have complete historical data at the first decision point.

¹⁵ The definition of $\widehat{W}_l^+$ can be found, e.g., DeMiguel et al. (2009b), Gu et al. (2020), and Kan et al. (2022).

¹⁶ In recent studies, MAXSER has been extended to accommodate daily return data characterized by heteroscedasticity and heavy tails (Ao et al., 2025), and to be applicable in high-dimensional settings where the cross-sectional dimension exceeds the time length (Chen et al., 2025).

¹⁷ In Section D of our supplementary material, we also report empirical results based on daily return data with investment pools consisting of components of the S & P 500 index or the Russell 2000 index, as well as monthly return data with investment pools consisting of 100 portfolios sorted by size and book-to-market and 49 industry portfolios, obtained from Ken French's website.

To avoid look-ahead bias due to missing data during the rolling window process, we simply drop the corresponding assets from the basket (Ao et al., 2019; Fan et al., 2024b). The risk-free data are downloaded from the Ken French's data library.¹⁸

Based on S & P 500 index (Russell 2000 index) components, we consider the following three scenarios: (1) the largest (smallest) $p = 50$ stocks (measured by the market values), (2) the largest (smallest) $p = 100$ stocks, (3) the largest (smallest) $p = 150$ stocks. The sample size T is fixed at 120 periods (ten years). The first two scenarios are suggested by one of the reviewers; these p and T combinations are commonly used in the portfolio allocation research, with p/T ratios of 0.42 and 0.83, respectively. In the third scenario, the number of assets p is greater than sample size T , and p/T ratio is 1.25. The target return $\bar{a}$ is set to 2%, and the corresponding target risk for MAXSER and SRM is set to $\bar{\alpha}/\sqrt{\hat{\theta}_{all}}$,¹⁹ and is determined at the first decision point, where $\hat{\theta}_{all} = \hat{\mu}^T \hat{\Sigma}_r^{-1} \hat{\mu}$. The long-term investment practices are considered, with the out-of-sample period spanning from 01/2000 to 12/2023. The portfolios are rebalanced monthly, meaning the holding period T_h is 1. In Section D.1 of our online supplementary material, we also present the empirical results when the stocks in the investment pool are randomly selected from the S & P 500 index or the Russell 2000 index components.

6.5. Results

Table 5 and 6 report the empirical results of all strategies under three scenarios based on S & P 500 index components and Russell 2000 index components, respectively. The proposed HD-DRO performs well in practical applications.

First, the HD-DRO exhibits an outstanding return–risk balance and is the best performer in terms of the Sharpe ratio in most cases. The SRs of HD-DRO are the highest across all scenarios when the investment pool consists of the largest S & P 500 index components. For example, in the case of $p = 100$, the SR of HD-DRO is 0.194, which is much higher than 0.145 of DRO, and the second and third highest SRs are achieved by RSP and R-MVP, respectively. Second, the HD-DRO takes on low out-of-sample risk in all cases, with its standard deviation often ranking among the top three lowest across all considered strategies. The lowest risk is often obtained by R-MVP, which is reasonable given that it is the robustified global minimum variance portfolio. Furthermore, the HD-DRO also has the almost lowest maximum drawdown in most cases. For example, in the case of the smallest 100 stocks from the Russell 2000 index, HD-DRO achieves the lowest maximum drawdown of 0.409. Low risk and low maximum drawdown are desirable features of a “robust” portfolio. Third, the portfolio turnover of HD-DRO often ranks fourth to sixth among the 13 considered strategies, indicating that the newly proposed strategy is less affected by transaction costs than most competitors. Evidently, the lowest portfolio turnover is obtained by the $1/N$ which ranks the first. The amount of negative weights of HD-DRO often ranks sixth to eighth among 13 strategies (the equal weighting strategy has zero NS and ranks the first). Next, the maximum and minimum weight of HD-DRO often ranks fourth to sixth among all strategies, indicating that our robust portfolio is less prone to extreme allocations and achieves better diversification across all assets. Finally, our HD-DRO performs well when the number of assets p exceeds the sample size T , achieving the highest SRs among the available competitors in most cases. For example, when the asset pool is made up of the largest 150 S & P index component stocks, the new robust portfolio achieves the highest SR of 0.140 among seven available strategies.

Several other interesting findings are summarized below: (1) The performance of DRO is similar to that of the $1/N$, particularly in terms of the measurements of the weights. It is worth mentioning that the performances of DRO and $1/N$ are nearly identical when the investment pool consists of portfolios obtained from Ken French's library, see Section D.3 of our supplementary material for more details. (2) The MAXSER achieves a top-tier Sharpe ratio in most cases, particularly when daily data is applied, see Section D of our online supplementary material. (3) For mean–variance portfolio MV-POET without considering uncertainty, it undertakes much higher risk than HD-DRO, and while it may achieve higher cumulative returns in some cases, its SRs are always lower than those of HD-DRO. (4) The lowest SR is often observed for MV, which is expected due to the application of the sample covariance matrix in the high-dimensional setup. MV-POETS consistently outperforms the MV across all considered cases. We notice that, it is somewhat surprising that the Sharpe ratio of MV-POET is worse than that of MV in many cases. One on hand, our unreported simulation results indicate that the performance of MV-POET is closely related to the choice of $\bar{a}$, an inappropriately selected high target return can lead to substantially poor performance. On the other hand, MV-POET often suffers significant losses (corresponding to substantially larger maximum drawdown) when the historical data includes periods of financial crisis, which leads to poor overall performance.

In the Section D.1 of supplementary material, we also plot the cumulative out-of-sample excess return of HD-DRO, DRO, and $1/N$. It can be observed that during the 2008 financial crisis, both DRO and $1/N$ portfolios experienced a sharp decline in returns, whereas HD-DRO maintained much greater stability, demonstrating its robustness. Furthermore, the DRO and $1/N$ exhibit highly similar performance across most periods and cases, with their cumulative return curves often running parallel, suggesting that DRO applies an overestimated uncertainty size.

7. Conclusion

In this paper, we study distributionally robust portfolio allocation based on a factor model in high-dimensional situations. Specifically, we impose an uncertainty set, the region centered by empirical probability measure with radius δ , on the common

¹⁸ https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

¹⁹ Given return constraint level $\bar{a}$, the corresponding risk constraint level is $\bar{\alpha}/\sqrt{\varphi}$, where $\varphi = \mu^T \Sigma_r^{-1} \mu$, see (1.1) and (1.2) of Ao et al. (2019).

Table 5

Empirical results based on monthly data of largest S & P 500 index components. We report the cumulative excess return (CR), the standard deviation of the portfolio excess return (Risk), the Sharpe ratio (SR), the Maximum drawdown (MDD), the portfolio turnover (TO), the amount of negative weights (NS), maximum weights (MAX), and minimum weights (MIN). The out-of-sample period is from 01/2000 to 12/2023. Sample size is set to 120. “–” means that the corresponding method is inapplicable.

	CR	Risk	SR	MDD	TO	NS	MAX	MIN
$p = 50$								
HD-DRO	1.504	0.041	0.128	0.645	0.239	0.535	0.222	−0.117
DRO	1.341	0.039	0.120	0.640	0.072	0.068	0.072	−0.027
1/N	1.459	0.051	0.099	0.988	0.053	0.000	0.026	0.024
MV	1.119	0.058	0.067	1.091	0.731	1.615	0.330	−0.306
MV-POET	0.417	0.286	0.005	4.006	3.505	2.489	0.436	−0.475
MV-POETS	1.123	0.058	0.067	0.945	0.584	1.128	0.306	−0.235
MV-L1	1.094	0.056	0.068	1.038	0.605	1.195	0.315	−0.282
CP-LW	1.739	0.066	0.091	1.636	0.802	1.272	0.265	−0.218
CP-SF	1.431	0.074	0.067	1.768	0.353	0.168	0.112	−0.039
RSP	1.240	0.045	0.097	1.024	0.379	0.369	0.145	−0.098
BonaFide	1.603	0.082	0.068	1.898	1.017	1.540	0.309	−0.257
R-MVP	1.321	0.040	0.116	0.652	0.165	0.380	0.196	−0.082
MAXSER	1.887	0.060	0.109	0.666	1.176	0.990	1.052	−0.628
SRM	1.355	0.048	0.099	0.936	0.561	0.547	0.322	−0.213
$p = 100$								
HD-DRO	2.091	0.038	0.194	0.472	0.303	0.703	0.148	−0.077
DRO	1.591	0.038	0.145	0.587	0.065	0.053	0.036	−0.015
1/N	1.659	0.050	0.117	0.820	0.052	0.000	0.014	0.013
MV	1.717	0.071	0.084	1.666	1.544	2.964	0.360	−0.331
MV-POET	−2.580	0.203	−0.044	4.044	2.603	2.233	0.254	−0.256
MV-POETS	1.458	0.058	0.088	1.165	0.659	1.309	0.203	−0.150
MV-L1	1.809	0.056	0.112	1.127	0.824	1.095	0.250	−0.174
CP-LW	2.471	0.061	0.141	1.184	0.779	1.416	0.183	−0.145
CP-SF	1.859	0.077	0.085	1.545	0.281	0.133	0.064	−0.024
RSP	2.161	0.040	0.186	0.574	0.481	0.622	0.113	−0.082
BonaFide	1.827	0.101	0.063	1.985	2.038	2.987	0.349	−0.320
R-MVP	1.973	0.037	0.185	0.466	0.231	0.556	0.136	−0.065
MAXSER	2.800	0.056	0.173	0.429	1.730	0.948	1.135	−0.563
SRM	1.854	0.046	0.141	0.810	0.620	0.586	0.302	−0.187
$p = 150$								
HD-DRO	1.410	0.035	0.140	0.435	0.355	0.742	0.114	−0.068
DRO	–	–	–	–	–	–	–	–
1/N	1.901	0.051	0.130	0.851	0.054	0.000	0.009	0.008
MV	–	–	–	–	–	–	–	–
MV-POET	−0.575	0.144	−0.014	3.279	3.821	1.763	0.148	−0.171
MV-POETS	1.089	0.049	0.078	0.922	0.626	1.201	0.158	−0.101
MV-L1	1.402	0.050	0.098	0.898	0.871	1.316	0.234	−0.170
CP-LW	–	–	–	–	–	–	–	–
CP-SF	–	–	–	–	–	–	–	–
RSP	–	–	–	–	–	–	–	–
BonaFide	0.928	0.066	0.049	0.815	2.602	2.253	0.203	−0.181
R-MVP	1.352	0.035	0.136	0.428	0.288	0.639	0.106	−0.059
MAXSER	–	–	–	–	–	–	–	–
SRM	1.387	0.060	0.081	1.021	0.727	0.618	0.296	−0.178

factors which is low dimensional when considering the mean–variance portfolio allocation. The distributionally robust model is a min–max problem that is hard to solve. Thus, we reform it to a regularized optimization problem via a dual argument. The optimization problem involves two parameters δ and ρ , which measure the size of the uncertainty set and is the lowest acceptable portfolio return, respectively. Both parameters are crucial for the portfolio allocation. Then, we discuss the data-driven procedures for choosing each of them. The optimal choices of δ and ρ require knowledge of some unknown population quantities. Therefore, we provide further estimation procedures and demonstrate the corresponding asymptotic convergence. The Monte Carlo simulation shows that the estimated values of optimal δ and ρ are close to the corresponding oracle version, and our distributionally robust portfolio enjoys ideal out-of-sample performance in terms of Sharpe ratio. Finally, we conduct an empirical analysis based on component stocks of the S & P 500 index and Russell 2000 index from 01/2000 to 12/2023. Compared to various existing benchmarks, particularly the distributionally robust portfolio of Blanchet et al. (2022), the proposed high-dimensional distributionally robust mean–variance portfolio exhibits superior performance in terms of Sharpe ratio and risk measures.

Table 6

Empirical results based on monthly data of smallest Russell 2000 index components. We report the cumulative excess return (CR), the standard deviation of the portfolio excess return (Risk), the Sharpe ratio (SR), the Maximum drawdown (MDD), the portfolio turnover (TO), the amount of negative weights (NS), maximum weights (MAX), and minimum weights (MIN). The out-of-sample period is from 01/2000 to 12/2023. Sample size is set to 120. “–” means that the corresponding method is inapplicable.

	CR	Risk	SR	MDD	TO	NS	MAX	MIN
<i>p</i> = 50								
HD-DRO	2.754	0.053	0.181	0.576	0.153	0.249	0.254	−0.048
DRO	2.899	0.056	0.180	0.634	0.112	0.058	0.087	−0.022
1/ <i>N</i>	3.494	0.068	0.178	0.650	0.097	0.000	0.028	0.025
MV	1.863	0.072	0.091	0.707	0.377	0.560	0.302	−0.128
MV-POET	5.268	0.118	0.156	1.097	0.257	0.232	0.109	−0.101
MV-POETS	1.859	0.065	0.099	0.653	0.235	0.363	0.248	−0.083
MV-L1	1.787	0.070	0.089	0.700	0.341	0.476	0.296	−0.120
CP-LW	3.533	0.086	0.143	0.597	0.569	0.741	0.228	−0.120
CP-SF	4.174	0.077	0.189	0.669	0.300	0.113	0.077	−0.020
RSP	2.434	0.056	0.150	0.632	0.237	0.096	0.117	−0.028
BonaFide	4.850	0.109	0.155	0.696	0.715	0.797	0.197	−0.128
R-MVP	2.706	0.052	0.182	0.583	0.138	0.199	0.236	−0.041
MAXSER	6.184	0.129	0.167	1.460	1.472	2.116	2.292	−1.366
SRM	4.121	0.106	0.136	1.018	0.774	0.753	0.442	−0.308
<i>p</i> = 100								
HD-DRO	2.549	0.040	0.222	0.409	0.223	0.385	0.231	−0.036
DRO	3.345	0.051	0.227	0.548	0.097	0.042	0.033	−0.012
1/ <i>N</i>	3.799	0.062	0.212	0.585	0.087	0.000	0.013	0.010
MV	0.267	0.080	0.012	1.125	1.301	1.785	0.320	−0.181
MV-POET	5.016	0.092	0.189	0.715	0.267	0.282	0.062	−0.077
MV-POETS	1.299	0.050	0.090	0.731	0.263	0.510	0.164	−0.056
MV-L1	0.951	0.060	0.055	0.769	0.658	0.913	0.252	−0.113
CP-LW	2.881	0.051	0.199	0.435	0.368	0.514	0.113	−0.058
CP-SF	3.929	0.066	0.207	0.572	0.175	0.021	0.030	−0.004
RSP	2.960	0.048	0.217	0.546	0.252	0.181	0.061	−0.026
BonaFide	5.502	0.088	0.217	0.904	1.234	1.399	0.216	−0.156
R-MVP	2.508	0.040	0.221	0.437	0.153	0.275	0.197	−0.027
MAXSER	5.849	0.100	0.204	0.840	1.923	2.257	1.834	−1.380
SRM	4.523	0.085	0.186	0.833	0.956	0.863	0.401	−0.267
<i>p</i> = 150								
HD-DRO	2.040	0.037	0.190	0.459	0.240	0.427	0.148	−0.038
DRO	–	–	–	–	–	–	–	–
1/ <i>N</i>	3.707	0.062	0.208	0.607	0.086	0.000	0.008	0.007
MV	–	–	–	–	–	–	–	–
MV-POET	5.151	0.093	0.192	0.659	0.379	0.385	0.055	−0.067
MV-POETS	0.908	0.044	0.072	0.644	0.283	0.585	0.119	−0.046
MV-L1	0.938	0.054	0.061	0.425	0.771	1.048	0.200	−0.111
CP-LW	–	–	–	–	–	–	–	–
CP-SF	–	–	–	–	–	–	–	–
RSP	–	–	–	–	–	–	–	–
BonaFide	4.222	0.074	0.199	0.667	1.418	0.903	0.097	−0.082
R-MVP	1.954	0.037	0.185	0.465	0.184	0.348	0.136	−0.031
MAXSER	–	–	–	–	–	–	–	–
SRM	3.022	0.077	0.136	0.560	0.991	0.873	0.316	−0.223

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jeconom.2025.106118>.

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